Did Expert Iteration Create It?

A Pretraining Knockout and Compute-Matched Budgets in a From-Scratch Theorem Prover

Dan Pandori et al. (author list for Dan to decide)

Draft, October 13, 2026. Not for circulation.

Abstract

Whether reinforcement learning (RL) against a verifier creates capabilities or raises the probability of behaviour the base already had is contested; large-model studies can rarely change what the base saw. We study it in a 9.56M-parameter transformer prover trained from scratch on 155,000 synthetic propositional-logic proofs and judged only by Lean 4. We remove one inference composition from pretraining, the classical reductio σ (assume $\neg G$, refute it, eliminate the double negation), keep both of its rules (each three to four times rarer), refill the data to the same size, and run 16 rounds of expert iteration (EI), a rejection-sampling form of RL. No knockout seed acquired a classical theorem family (half its held-out members proved in 256 attempts): 0/6 against 6/6 full-exposure seeds, a pre-registered contrast ($p = 0.00024$) largely present before RL. During RL, full-exposure seeds gained 8–69 held-out classical theorems and knockout seeds 0–2; 5/6 against 0/6 acquired a new family ($p = 0.0032$; not pre-registered). Restoring σ in 1% of pretraining restored acquisition (3/3 seeds; exploratory). The knockout bases were not σ-free: five of six proved held-out theorems with σ, mostly in one family, 1–7 times in 1.1–1.5 million attempts. EI practised σ on easier theorems but did not carry it to those that need it. In three earlier full-exposure runs, 51–60 theorems per run look RL-created at equal sample counts; 14–21 remain when the base gets RL's compute as attempts. In this organism EI amplified a composition pretraining had exposed, but within 16 rounds did not acquire the classical families without it.

1 Introduction

Reinforcement learning against a verifier is now the standard way to improve reasoning models. Whether it teaches them anything new is contested. Yue et al. [59] find that base models overtake their RL-trained descendants in pass@k as k grows. Karan and Du [21] approach RL's gain on a mathematics benchmark by sampling the base from a sharpened distribution, with no training. Other work reports RL solving problems the base never solved [1, 28]. Part of the disagreement is about budgets. A base whose per-attempt success probability on a problem is 10^{-6} will almost never solve it in 1,024 attempts, yet the capability is present in a sense that matters for evaluation and safety. A capability is defined only relative to a sampling budget [5, 27].

A claim that RL *created* a capability therefore needs three things:

- a budget at which the base is judged unable, tied to what RL spent;
- controls that receive RL's training without RL's proofs;
- ideally, a base that never had the capability [9, 18].

Interventions on what the base saw exist, and they disagree. Zhang et al. [61] vary a context's share of pretraining (0%, 0.1%, 1%) in a 100M model and find no transfer from RL below 1%.

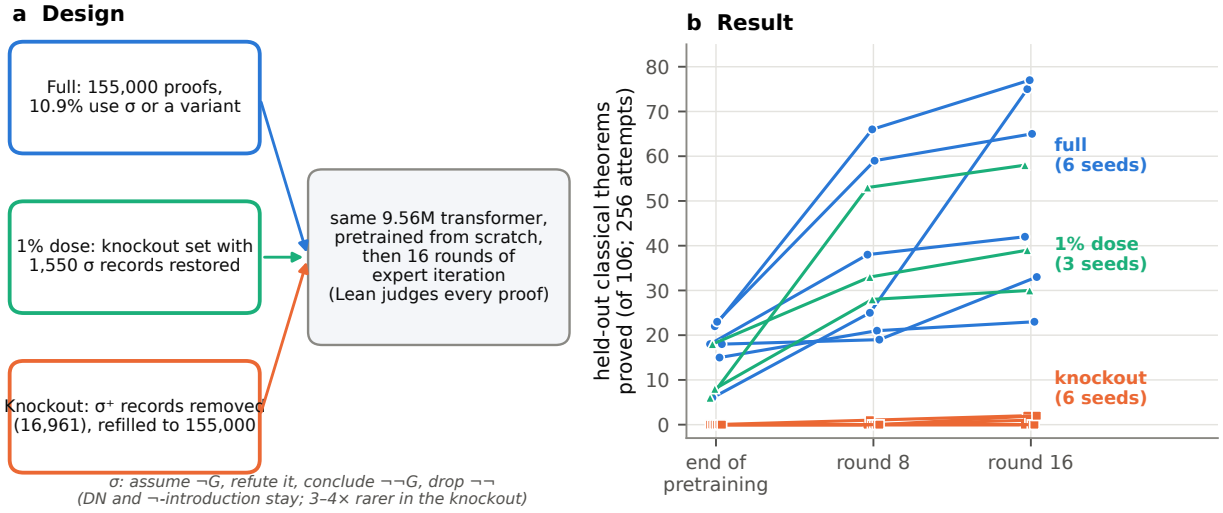


Figure 1: Design and result. (a) Three pretraining sets that differ in their σ^+ records (σ and its variants). (b) Held-out strict theorems proved (of 106, with 256 attempts each) at the end of pretraining and after 8 and 16 rounds of expert iteration, one line per seed. No knockout seed acquires a classical family in 16 rounds.

Abdulsalam et al. [1] exclude shortcuts from pretraining and see GRPO build them. Tsilivis et al. [46] set the share of long demonstrations exactly. A strategy removed at the fine-tuning stage did not re-emerge under GRPO [24], while a pattern family removed from the RL data did [31]. None of these measures the removed behaviour's rate under the base. We remove a single inference composition from pretraining and refill the data to the same size. Both of the composition's rules stay in the data, though each becomes three to four times rarer. We then measure the base's rate for the composition at about a million attempts.

We do this in a model organism small enough to allow it. It is a 9.56M-parameter transformer prover for propositional logic, pretrained from scratch on 155,000 synthetic natural-deduction proofs. It acts through a proof-state interface, one inference step at a time, and Lean 4 alone decides whether a proof counts. The composition we remove is the classical reductio σ: to prove a positive goal G, assume $\neg G$, derive a contradiction, conclude $\neg\neg G$ by negation introduction, and eliminate the double negation (`DN`, Lean's `Classical.byContradiction`). We track six classical theorem families, such as excluded middle and Peirce's law, whose *strict* members can only be proved by eliminating a double negation the proof itself built. Their 106 held-out strict theorems are never trained on. We then run expert iteration (EI) from each base for 16 rounds: sample, keep what Lean accepts, fine-tune (Fig. 1).

Two objections, and the null model. The first objection is that a model cannot learn what it never saw. But the knockout bases saw both rules, and five of six still proved held-out strict theorems with σ at large sample sizes. The second objection is stronger. At the bases' measured rates, EI's 3,936 attempts per round on the 123 strict training theorems should find a σ proof only rarely: at most about 0.02 to 0.03 per round. We take this waiting-time arithmetic as the null model and ask, post hoc, what happened beyond it.

Results.

- **No knockout seed acquired a classical family in 16 rounds** (0/6, which bounds the per-seed rate below 0.39), against 6/6 full-exposure seeds. That pre-registered contrast is largely decided before RL, because one family is already acquired at the end of pretraining on 5/6 full seeds. The RL-phase contrasts were not pre-registered. During the RL phase, which includes replay of pretraining data, full seeds gained 8–69 strict held-out theorems and knockout seeds 0–2 (exact Mann–Whitney $p = 0.0011$; the same in guided evaluations). On the three full seeds that shared the knockout's hardware, $p = 0.012$. Five of six full seeds against none acquired a new family ($p = 0.0032$).
- **One percent is enough.** Restoring σ in 1% of pretraining records, about a tenth of the full set's share, restored acquisition on 3/3 seeds (exploratory).
- **Rare, not absent, mostly in one family.** The knockout bases' successes at large k are almost all on converse contraposition, the family the full bases had before RL. On the other five families they wrote one σ proof in 6.7 million attempts. With guided decoding, four of six knockout bases proved a strict theorem in 256 attempts, including the only base whose rate is certified out of reach under plain sampling.
- **Beyond the null** (post hoc). The knockout ladders wrote σ from round 1 on theorems that supply the double negation. Three of six reached a strict training theorem sooner than their base rates predict: 0.55–0.74 seeds expected by round 16 against 3 observed ($p = 0.006$–0.019 under the null). None carried the step to a family. The ladder that went furthest, continued to 24 rounds, proved the same 5 of the 106 held-out theorems at rounds 20 and 24, more often each time.
- **Measurement.** In three earlier full-exposure runs, 51–60 theorems per run look RL-created at equal sample counts. Only 14–21 remain when the base gets RL's compute as attempts, under every budget allocation we tried.

Contributions.

1. **A composition knockout in pretraining** (§3), with a dose arm and diagnostics that separate "absent" from "rare": base rates at about 10^6 attempts, certificates, and likelihoods of known proofs along every RL run. The primary indicator, the certificates and the likelihood expectations were pre-registered, the dose arm by a dated amendment. The RL-phase contrasts and the waiting-time analysis are post hoc. An analyst who had not written the experiment's code recounted it independently.
2. **A compute-matched measurement protocol** (§2, §4), applied to three earlier full-exposure runs, and at one allocation to the knockout. It gives the base a budget of attempts matched to RL's compute under several defensible allocations, so that the "created" set becomes a curve in the budget. It brackets each theorem's success rate between sampling bounds and a likelihood estimate, certifies unreachability, and nets out RL-free training. It builds on budget-relative definitions [8, 27] and on base-to-RL exchange rates [44].
3. **An open organism**: a from-scratch prover with a sound judge, likelihoods of arbitrary proofs at every checkpoint, and pretraining that can be edited one composition at a time. Code, data, checkpoints and per-arm compute are public.

What transfers. The numbers do not: the 1% threshold, the 10^{-6} rates and the dynamics of EI belong to this organism. Three findings may.

- **The waiting time scales with kNr.** Here k is attempts per problem, N the number of training problems that need the composition (here 123 of 4,495), and r the base's rate. The null is a property of the RL pool as much as of pretraining.

- **Cued practice did not transfer.** Each knockout ladder wrote thousands of accepted proofs using σ or a variant where the problem supplied the double negation, and almost none where it did not.
- **Exposure set what EI could find, not what the model could learn.** In this organism, 16 demonstrations taught excluded middle to a base pretrained with no `DN` at all.

Failure at modest k does not show absence [5, 15]. Our knockout bases fail at $k = 256$ under plain sampling, yet guided decoding or a million plain attempts reveals the composition.

2 The organism and the experiment

Logic, data and judge. Theorems are sequents $\Gamma \vdash G$ of propositional logic over at most four atoms, with $\wedge, \vee, \rightarrow, \neg, \bot$. Proofs are Fitch-style natural deduction: premises, assumptions that open boxes, the introduction and elimination rules of each connective, $\bot$-elimination and double-negation elimination (`DN`). The pretraining set $\mathcal{D}$ ("K12" in the released code) holds 155,000 synthetic proofs of 2–12 lines from a generator, with a flat length distribution. A proof counts if and only if Lean 4 core accepts it. A syntactic pre-filter may reject malformed text before Lean, but never accepts anything on its own (§3.8).

Model and interface. Every model is the same decoder-only transformer: 6 layers, width 384, ALiBi and NoPE attention heads, 9,560,832 parameters. It is pretrained from scratch for 1,200 s on one NVIDIA A40, about 24,000 steps or 3.7 passes over the (state, action) pairs of $\mathcal{D}$. It acts in a *proof-state environment*. It sees the focused goal's Lean tactic state and writes one step, such as `have n5 : ¬P := n1 n4`. The environment applies the step, opens and closes the boxes of $\rightarrow$- and $\neg$-introduction and the branches of $\vee$-elimination, and ends the attempt at the first step it cannot apply. It also numbers the lines from one of 33 random starting offsets, so every proof has 33 spellings. The finished proof is rendered in Lean and judged.

Expert iteration. Each round, the current model samples $k = 32$ attempts at temperature $T = 0.8$ on each of 4,495 training theorems. Accepted proofs accumulate across rounds, up to four per theorem. The model is then fine-tuned for 600 steps, from where it left off, on those proofs (each repeated four times) mixed with 20,000 records replayed from its own pretraining set. The training budget is the same for every theorem. This is rejection-sampling fine-tuning in the lineage of expert iteration [3, 34], STaR [60] and ReST^{EM} [43]. We call one 16-round run from one pretrained seed a *ladder*, and the model after N rounds rN; r0 is the pretrained base. Each round also samples $k = 32$ attempts on every held-out family member, which is never trained on.

Evaluation. An evaluation draws $k = 256$ independent attempts per theorem at $T = 0.8$, with at most 96 steps and 512 tokens per step. A theorem is *proved* if any attempt is accepted. Two sample seeds give two independent draws of the same checkpoint. A *guided* evaluation uses the same budget, but a step that a per-step logical checker rejects is redrawn from the same state without replacement, up to 10 times per attempt. The checker never rejected a step Lean accepts in 314,541 tested (state, step) pairs. General ability is measured on 5,000 held-out generator theorems decoded greedily, and on 322 evaluation theorems (72 textbook problems plus 250 held-out generator theorems; §4).

The composition and the theorem families. Table 1 defines the terms used throughout (glossary: Appendix A). In $\mathcal{D}$, σ occurs in 16,703 records and σ^{+} in 16,961 (10.94%); the lemma

Table 1: The terms that separate "absent" from "rare". A *derived* line is one produced by a rule, not a premise or an assumption.

term	definition
σ (classical reductio)	`DN` applied to a $\neg\neg G$ that the proof built by $\neg$-introduction (assume $\neg G$, derive $\bot$). In Lean: `Classical.byContradiction` on that refutation.
σ^+	`DN` applied to any derived line (σ plus routes through $\to$-introduction, $\vee$-elimination, …).
lemma route	prove $\neg\neg G \to G$ inside an implication box (`DN` on the box's assumption), then apply it to a $\neg\neg G$ built by $\neg$-introduction. Classical, but not σ^+.
free positive reductio	a σ^+ step whose conclusion G is not a negation and whose $\neg\neg G$ does not occur in the theorem. It is the genuinely classical use; using σ on a theorem that supplies $\neg\neg G$ is not.
classical-only	the theorem has no intuitionistic proof (decided by the complete G4ip procedure).
strict member	a classical-only family member with no $\neg\neg$ subformula in its sequent. Every proof of it must eliminate a double negation that the proof built: by a free positive reductio or by the lemma route.

Table 2: Strict members per family. The held-out total adds one generated theorem, `la_transfer_916`.

	conv. contrap.	neg. cond.	Peirce	Peirce seq.	excl. middle	De Morgan	total
training (of 40 each)	27	16	15	18	38	9	123
held-out (of 40 each)	18	15	12	19	37	4	105 + 1

route occurs in none. We track six classical generator families: converse contraposition ($\neg B \to \neg A \vdash A \to B$), negated conditional ($\neg(A \to B) \vdash A \wedge \neg B$), Peirce's law, the Peirce sequent ($((A \to B) \to A \vdash A$), excluded middle, and De Morgan ($\neg(A \wedge B) \vdash \neg A \vee \neg B$). Each has 40 training members and 40 held-out members, substitution instances of the family's schema. The strict members number 123 in training and 106 held out: 105 family members plus one generated theorem, `la_transfer_916` (Table 2). A model *acquires* a family if it proves at least half of the family's strict held-out members in one evaluation. Figure 2 shows a σ proof written by a knockout base.

Arms. The arms share the network, the pretraining recipe and the EI protocol (Table 3).

- **knockout** pretrains on $\mathcal{D}^-$. This is $\mathcal{D}$ minus exactly its 16,961 σ^+ records, refilled with 16,961 records drawn length by length from $\mathcal{D}$'s own unused source pools. Every refill record passes $\mathcal{D}$'s filters and contains no σ^+ and no lemma route. Under two renaming-invariant keys, the refill shares no theorem with $\mathcal{D}$ or with any evaluation file. Lean accepts all 16,961 refill records (Appendix B).
- **dose** restores 1,550 removed σ records in place of refill records, at $\mathcal{D}$'s length mix: 1.00% of the records.
- **full** pretrains on $\mathcal{D}$. Seeds s0–s2 come from earlier runs. They differ from the new seeds s3–s5 in three ways. Rounds 1–8 ran on RTX A6000 cards at sampling batch 2,048 (s2 at 1,024 from round 2), whereas the new seeds ran on A40s at 1,024. The ladders were resumed at rounds 9 and 13, which restarted the replay sampler. And each round's evaluation phase sampled more theorems. Changing the batch changes which samples are drawn, not how they are judged.

Because each ladder replays its own pretraining data, exposure differs during RL as well. Full

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬P))) (h2 : P) : (R → S) := by
  have n1 : ((¬S) → (¬P)) := h1
  have n2 : P := h2
  have n9 : (R → S) := (fun (n3 : R) => by
    have n7 : (¬(¬S)) := (fun (n4 : (¬S)) => by
      have n5 : (¬P) := n1 n4
      have n6 : False := n5 n2
      exact (n6 : False))
    have n8 : S := Classical.byContradiction (fun hh => n7 hh)
    exact n8)
  exact n9
```

Figure 2: A free positive reductio written by a knockout base before any RL: $\neg S \to \neg P,\ P \vdash R \to S$ (`la_transfer_916`), one of the rare successes of knockout seed 1's base in large-k sampling. Assume $\neg S$ and refute it (`n7`), then eliminate the double negation (`n8`). Lean accepts it and rejects the same proof of the negated goal. Appendix E gives two more.

Table 3: Arms. Every model: 9,560,832-parameter transformer, proof-state interface, pretrained from scratch, then 16 rounds of expert iteration ($k = 32$, $T = 0.8$, 600 fine-tuning steps per round, 4,495 training theorems). Lean 4 alone judges.

arm	pretraining set (155,000 records each)	seeds	σ^+ share
FULL	$\mathcal{D}$	6 (s0–s5)	10.94%
KNOCKOUT	$\mathcal{D}^-$: $\mathcal{D}$ − 16,961 σ^+ records + 16,961 refill	6 (s0–s5)	0
DOSE	$\mathcal{D}^{1\%}$: $\mathcal{D}^-$ with 1,550 σ records restored	3 (s0–s2)	1.00%

ladders replay about 2,200 σ^+ records per round, dose ladders about 200, and knockout ladders none. "Exposure" below therefore means pretraining together with the replay drawn from it. Arms with the same seed number share initial weights and sampling seeds, not data.

Budgets, reach and certificates. We write $p_M(t)$ for the probability that one attempt of model M on theorem t is accepted, and $\hat{p} = c/n$ for its estimate from c successes in n attempts. For a budget of K attempts per theorem:

$$\begin{aligned} t \text{ within reach of the base at } K &\iff \hat{p} \geq 1/K, \\ t \text{ not reached at } K &\iff c = 0 \text{ with } n \geq K, \\ t \text{ certified out of reach at } K &\iff \mathrm{UB}_{95}(c, n) < 0.05/K, \end{aligned}$$

where UB_{95} is the one-sided 95% Clopper–Pearson bound; with $c = 0$, certification needs $n \approx 60K$. "Within reach" from a single success in fewer than K attempts is weak evidence, and we flag it where it occurs. A *pooled* certificate over a set of theorems bounds their mean per-attempt rate, not any one of them. K is matched to RL's compute as $K = C_{\mathrm{RL}}/(N_{\mathrm{eval}}\, c_{\mathrm{base}})$. Here C_{RL} is what the ladder spent, and c_{base} is the base's measured cost per attempt. "What the ladder spent" has several defensible readings, each an *allocation*: its sampling attempts on training theorems (K_{att}); the GPU time of its sampling and fine-tuning phases (K_{train}; $K_{\mathrm{train}}(16)$ for a 16-round ladder); all of its GPU time (K_{uni}); the value of an earlier report (K_{pub}, §4); or an adaptive spend of 512 attempts on every theorem and the rest on the base's failures (K_{adapt}). $N_{\mathrm{eval}} = 322$ is the number of evaluation theorems the budget is spread over. N_{eval} is a choice: spreading the same compute over the 1,054 theorems of our largest evaluation would make every K 3.3 times smaller. The

Table 4: Strict held-out theorems proved (of 106) per seed. r0 is the base. x0 and x1 are two sample seeds of the plain evaluation; g is the guided evaluation. Families: cc = converse contraposition, nc = negated conditional, pe = Peirce, ps = Peirce sequent, em = excluded middle; underlined = already acquired at r0. † = seeds from earlier runs. The dose arm's other r16 evaluations are in Appendix I.

model	r0 x0	r0 g	r8 x0	r16 x0	r16 x1	r16 g	acquired at r16 (x0)	gain r0→r16 (x0)
FULL s0†	18	20	38	42	43	44	cc, nc	24
FULL s1†	6	13	25	75	73	75	cc, em, pe	69
FULL s2†	22	36	66	77	77	80	cc, nc, pe, ps	55
FULL s3	23	35	59	65	64	64	cc, nc, pe, ps	42
FULL s4	15	19	21	23	22	24	cc	8
FULL s5	18	17	19	33	34	33	cc, ps	15
KNOCKOUT s0	0	1	0	0	0	0	–	0
KNOCKOUT s1	0	1	1	2	2	3	– (r20: 5; r24: 5)	2
KNOCKOUT s2	0	0	0	1	1	0	–	1
KNOCKOUT s3	0	1	0	0	0	0	–	0
KNOCKOUT s4	0	3	0	0	0	0	–	0
KNOCKOUT s5	0	0	0	2	1	2	–	2
DOSE s0	6	8	53	58			cc, nc, pe, ps	52
DOSE s1	18	21	33	39			cc, nc	21
DOSE s2	8	13	28	30			cc, nc	22

compute-matched set $\mathcal{C}(K)$ holds the theorems that RL proves and its base cannot reach within K. $\Delta_{\text{cov}}(K)$ is the number RL proves minus the number the base reaches.

The pre-registered decision rule. Per family, at r16:

- CREATED if at least 2 of 6 knockout seeds acquire it and the knockout base is certified out of reach on it;
- NEEDS EXPOSURE if at most one knockout seed acquires it and more full seeds do (one-sided Boschloo test, $p \leq 0.05$);
- NEITHER if no seed acquires it;
- UNDETERMINED otherwise.

The primary test compares, per seed, the indicator "at least one family acquired at r16" between the full and knockout arms. The pre-registration fixed the certificate pool as the 106 strict held-out theorems, at each knockout seed's training-phase budget $K_{\text{train}}(16)$. As sampled, about 10^4 attempts per theorem, no single family could be certified at that budget, so CREATED could not be returned. Section 3 reports what the remaining verdicts can and cannot say.

3 Results: the σ knockout

All numbers in this section are on the 9.56M-parameter models of Table 3, judged by Lean alone. They come from the evaluation with sample seed x0 unless stated. An analyst who had not written the experiment's code recounted the experiment's counts from the raw files, with separate code, and re-checked the proofs in Lean. Numbers computed for this paper from those recounted files are marked (*this paper*).

3.1 No knockout seed acquired a family in 16 rounds

No knockout seed acquired any classical family by r16, and every full seed had at least one (Table 4, Fig. 3). This is the pre-registered primary indicator: 6/6 against 0/6, one-sided Boschloo $p = 0.00024$. The same holds in the second sample seed, in the guided evaluation, and with the looser held-out sets first pre-registered (178 theorems). Those were replaced by the strict sets in a dated amendment after round 1 of the first knockout ladder, because that base proved 136 of 155 generated "classical" targets using a $\neg\neg$ that the sequent supplied. The 0/6 bounds the per-seed rate at which this EI acquires a family from a σ-free pretraining set below 0.39 (one-sided 95%).

The indicator is a weak statement about RL. Converse contraposition is acquired *before* RL on 5/6 full bases (6/6 in the guided evaluation). Applied to the bases, the indicator already gives 5/6 against 0/6 ($p = 0.0032$; 6/6 against 0/6 guided, $p = 0.00024$; *this paper*). Per family, the pre-registered rule gives NEEDS EXPOSURE only for converse contraposition (6 vs 0 seeds; Holm-corrected over six families, 0.0015). It gives UNDETERMINED for negated conditional, Peirce and the Peirce sequent (3 vs 0 each; $p = 0.034$, Holm 0.17) and for excluded middle (1 vs 0). It gives NEITHER for De Morgan, whose four strict members no model proves.

The RL-phase comparisons were not pre-registered. The RL phase includes replay of pretraining data, so they compare the whole RL phase, not RL proofs alone. During it, the full seeds gained 24, 69, 55, 42, 8 and 15 strict held-out theorems and the knockout seeds 0, 2, 1, 0, 0 and 2. The arms do not overlap: exact one-sided Mann–Whitney $U = 36$, $p = 0.0011$ (*this paper*). In the guided evaluations the gains are 5–62 against -3 to $+2$, again $p = 0.0011$. On the three full seeds that ran on the knockout's hardware, $p = 0.012$. Five of six full seeds acquired a family during RL and no knockout seed did ($p = 0.0032$). By coincidence this is the same 5/6 as the r0 indicator, from different seeds. On the three new full seeds alone it is 2/3 against 0/6 ($p = 0.034$). On the training side, the knockout ladders proved none of the 123 strict training theorems in round 1 and only 0, 5, 1, 0, 0 and 2 of them by round 16. The full ladders went from 21, 1, 27, 17, 15 and 20 in round 1 to 49, 80, 86, 78, 33 and 36 (Fig. 4a). No knockout ladder ever had proofs for half of any family's strict training theorems (Appendix C).

3.2 One percent is enough

The dose bases proved 6, 18 and 8 strict held-out theorems at r0, and 58, 39 and 30 at r16. All three acquired families during RL; dose s1 had already acquired converse contraposition at r0. That is 3/3 against the knockout's 0/6 (Boschloo $p = 0.0033$). This arm was added by a dated amendment before it ran, and it is exploratory. At r16 the dose seeds lie inside the full arm's range (two-sided Mann–Whitney $p = 0.55$ with three seeds; *this paper*). The dose restores about a tenth of the full set's σ^+ share (1.00% against 10.94%), and it moves the constituents little. In $\mathcal{D}^{1\%}$, `DN` appears in 4.24% of records and $\neg$-introduction in 6.12%, against 3.29% and 5.19% in $\mathcal{D}^-$ and 13.73% and 15.21% in $\mathcal{D}$. Classical-only theorems are 1.56% of a 20,000-record sample, against 1.35% and 4.17% (*this paper*, from the dose set's file). In round 1 the dose bases wrote 223, 183 and 241 σ proofs, the new full seeds 635–885 and the knockout seeds 6–13. The dose bases' rates were read only at $k = 256$ and in round 1, not at large k. A second sample seed and a guided evaluation, run for this paper, prove 56, 39 and 30 (second seed) and 58 and 42 on seeds 0 and 1 (guided; seed 2 pending), with the same families (Appendix I).

3.3 σ is rare in the knockout arm, not absent

The bases, sampled about a million times. We drew 1.1–1.5 million attempts from each knockout base, spread over the 106 strict held-out theorems (about 10^4 per theorem, with sampling

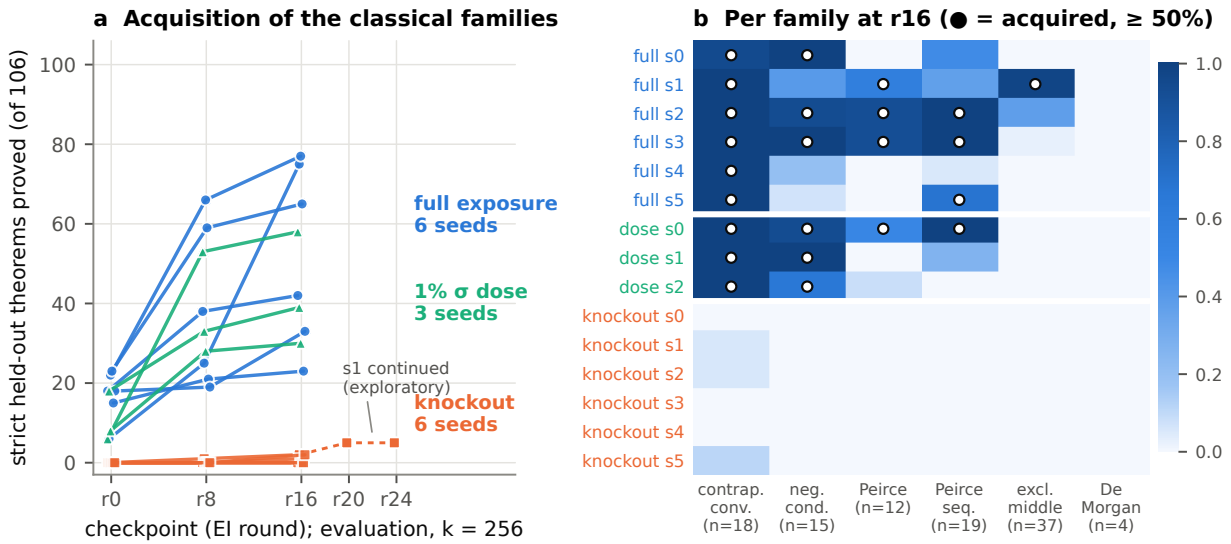


Figure 3: Per seed and per family. (a) Strict held-out theorems proved at r0, r8 and r16; knockout seed 1 was continued to r24 (dashed; exploratory). (b) Share of each family's strict held-out members proved at r16; a dot marks an acquired family ($\geq 50\%$).

Table 5: Knockout bases at large k (plain sampling, $T = 0.8$), pooled over the six bases per family (*this paper*, from the independent recount's files). UB = one-sided 95% Clopper–Pearson bound on the pooled rate.

family (strict held-out members)	attempts	successes	UB	successes by seed (s0–s5)
converse contraposition (18)	1,349,310	11	1.35×10^{-5}	0, 6, 1, 0, 3, 1
Peirce sequent (19)	1,459,630	1	3.2×10^{-6}	0, 0, 0, 0, 0, 1
excluded middle (37)	2,862,986	0	1.05×10^{-6}	–
negated conditional (15)	1,160,670	0	2.6×10^{-6}	–
Peirce (12)	928,536	0	3.2×10^{-6}	–
De Morgan (4)	309,512	0	9.7×10^{-6}	–
`la_transfer_916` (generated)	53,872	2		0, 1, 0, 1, 0, 0

on a theorem paused after its first success in each block of 1,024). Five of the six bases proved strict theorems: seed 1 seven times, seed 4 three times, seed 5 twice, and seeds 2 and 3 once each. Every success is a genuine σ proof. The successes are concentrated (Table 5). Eleven are on converse contraposition, the family the full bases mostly had before RL, and two are on the generated theorem `la_transfer_916`. On the other five families the six bases together wrote one σ proof in 6.7 million attempts. Excluded middle had none in 2.9 million. Pooled rates per base are 0 to 6×10^{-6} per attempt. On the theorems with a success, the per-theorem estimates are 7.5×10^{-5} to 4.9×10^{-4}, each from one to three successes. With guided decoding, at about 2.6 times the cost per attempt, four of the six bases proved a strict theorem in 256 attempts at r0: seeds 0, 1, 3 and 4.

Certificates. As pre-registered, pooled over the 106 theorems at each seed's $K_{\text{train}}(16)$, only seed 0's base is certified out of reach. It had 0 successes in 1.50 million plain attempts: $\text{UB}_{95} = 1.99 \times 10^{-6} < 0.05/K = 2.14 \times 10^{-6}$ (Table 6). Its certificate holds for budgets up to $K_{\max} = 25{,}112$, 8% above the budget, and it holds for plain sampling only: its guided evaluation at r0 proved a

Table 6: Knockout bases at large k, pooled over the 106 strict held-out theorems (the pre-registered pool), at each seed's $K = K_{\text{train}}(16)$, for compute spread over the 322 evaluation theorems. Certified out of reach if $\text{UB}_{95} < 0.05/K$; $K_{\max} = 0.05/\text{UB}_{95}$ is the largest budget the data certify (*this paper*). In parentheses: the 105 family members only.

knockout base	K	attempts	successes	$K_{\max}$	certified
s0	23,322	1,504,600	0 (0)	25,112 (24,892)	yes (yes)
s1	17,189	1,122,426	7 (6)	4,268 (4,730)	no (no)
s2	20,907	1,350,143	1 (1)	14,230 (14,106)	no (no)
s3	20,782	1,342,688	1 (0)	14,152 (22,376)	no (yes)
s4	21,554	1,372,403	3 (3)	8,850 (8,772)	no (no)
s5	22,297	1,432,256	2 (2)	11,375 (11,275)	no (no)

converse-contraposition member. The number of certified bases depends on how the budget is spread (§2). With the same compute spread over the 106 strict theorems it is 0 of 6, over the 322 evaluation theorems 1 of 6, and over 1,054 theorems 5 of 6 (Appendix D). Pooled over the 105 family members alone, as the experiment reported it, seed 3 is also certified; the difference is its success on `la_transfer_916`. No family is certified for any seed, since the largest certifiable K per family is 717–8,772. A pooled certificate bounds the mean rate over the pool, not any theorem in it. It also assumes that the pre-filter never rejected a valid proof in the large-k sample, which we could not check (§3.8).

The ladders. Every knockout ladder wrote σ in round 1, 6–13 distinct proofs per ladder, on theorems that supply $\neg\neg G$ or whose goal is a negation. Over 16 rounds each wrote about 2,200–4,400 distinct accepted proofs using σ^+ (0.33–0.64% of its distinct proofs; *this paper*, from the recount's counts and shares), nearly all of that kind. Its fine-tuning mixes carried 16–276 σ records per round (each proof repeated four times; 844–3,272 records over rounds 1–16). The free positive reductio was rare. Over 16 rounds the knockout ladders produced 0, 34, 4, 1, 2 and 16 distinct proofs using it, first in rounds never, 9, 12, 15, 15 and 2. The full ladders produced 17,507–50,222 each, and the dose ladders 20,597–26,662, from round 1 (Fig. 4b). EI practised σ where the problem supplied the double negation, and did not carry it to where it did not.

Beyond the waiting-time null (*this paper*, post hoc). Suppose each knockout ladder kept sampling the 123 strict training theorems at its base's pooled rate, with held-out rates standing in for training ones. It would then expect a first strict success only after dozens to hundreds of rounds: 47 for seed 1 and 115–340 for seeds 2, 4 and 5, with plug-in rates on the 105 family members. The expected number of seeds with a strict success by round 16 is 0.55. Three had one: seed 1 in round 9, seeds 2 and 5 in round 14. Under the null that has probability 0.006. Under other reasonable choices the probability rises to 0.019, and the expected number of seeds to 0.74. Those choices are the 106 pool, family weights matched to the training mix (converse contraposition is 22% of the strict training theorems and 17% of the held-out ones), and a Jeffreys posterior on each rate. Either way the knockout ladders' rate rose during RL. Why it rose is not identified: practice on cued σ, general drift from fine-tuning, or both. It did not rise enough to acquire a family.

Likelihood. We scored the best known σ proof of each of 101 strict held-out theorems under every checkpoint, teacher-forced at $T = 0.8$ (Fig. 5). The proofs are the full ladders' own. At r0 the median log-probability is -15.4 to -20.9 nats under the four full bases scored, -18.9 to -25.3

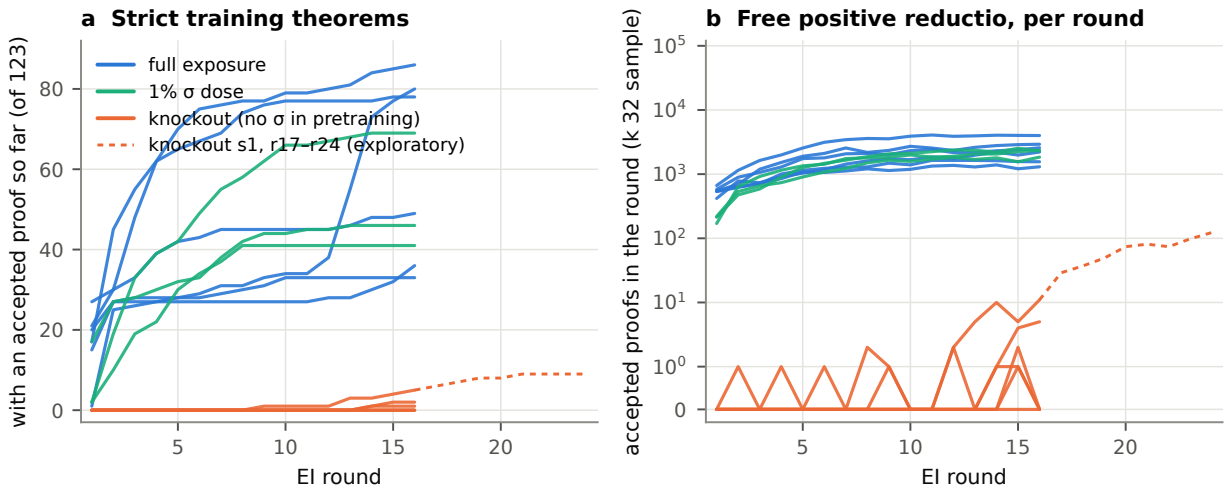


Figure 4: Ladder dynamics. (a) Strict training theorems (of 123) with an accepted proof so far. (b) Accepted proofs per round, in each round's $k = 32$ sample over all 4,495 training theorems, that use a free positive reductio (symmetric-log scale). Full and dose ladders write hundreds in round 1 and thousands later. Knockout seed 5 first writes one in round 2; the others write a few, late or never. Knockout seed 1 (dashed beyond round 16) spreads it, from 11 in round 16 to 123 in round 24.

under the dose bases, and −26.0 to −36.0 under the knockout bases. Full and dose overlap; the nearest dose and knockout bases are within 0.7 nats. The costliest step is the box that assumes $\neg G$: −18.0 to −21.6 nats under the knockout bases, −10.5 under full seed 0's base. Two ladders move strongly toward these proofs by r16 (full s3 +10.3 nats, dose s0 +16.6). The other full and dose ladders move +1.7 to +4.3, and the knockout ladders −7.6 to +3.7. The path matters as well. By r4, knockout seeds 2 and 3 had risen 8.5 and 8.1 nats, more than full seeds 4 and 5 (+5.0, +4.2) and dose seeds 1 and 2 (+1.6, +2.7), and then they fell back. Only the r16 endpoint separates the arms. Two caveats apply. The scores use one of the 33 line-numbering offsets, so they understate a proof's probability by at most $\ln 33 \approx 3.5$ nats. And a knockout ladder that found its own spelling (seed 1, below) need not raise these.

3.4 The one knockout ladder that moved

Knockout seed 1's free positive reductio kept spreading, so it was continued to round 24 (exploratory). Its free positive reductio proofs per round rose from 11 in round 16 to 123 in round 24. From round 21 it had proofs for 9 of the 27 strict converse-contraposition training theorems, and it never got past 9 strict training theorems in all. By the earlier organisms' criterion for igniting a pattern (at least 2% of the targets that need it), it ignited at round 13 (3 of 123), but it did not acquire a family. It proved the same five strict held-out theorems at r20 and r24, each more often: four converse-contraposition members and `la_transfer_916`. The largest gain is on `la_transfer_1019`, from 0 to 32 to 214 successes in 256 attempts at r16, r20 and r24. Its r20 proofs of these theorems take three routes: σ (32 proofs), other σ^+ (24) and the lemma route (7), a route absent from every pretraining set.

Is this elicitation of something the base already did? The evidence is mixed. In the large-k sample the base proved four of its 18 converse-contraposition members, one to three times each in 7,168–10,240 attempts. That is within reach by the rate definition, but always from fewer than K attempts. Three of the four members the ladder proves (`1246`, `1460`, `2256`) are among them,

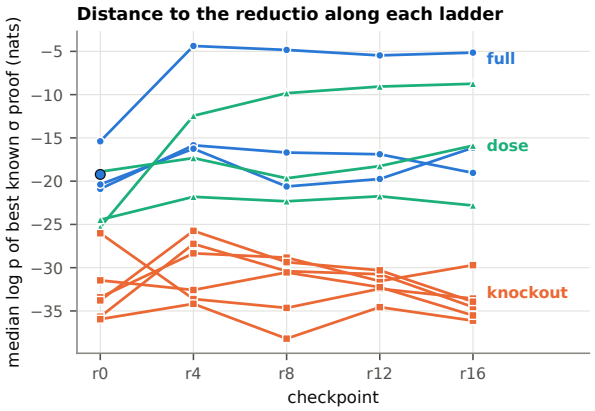


Figure 5: Median log-probability of the best known σ proof of the strict held-out theorems, along each ladder (one line-numbering offset, $T = 0.8$). The outlined point is full seed 0's base, whose ladder was not rescored.

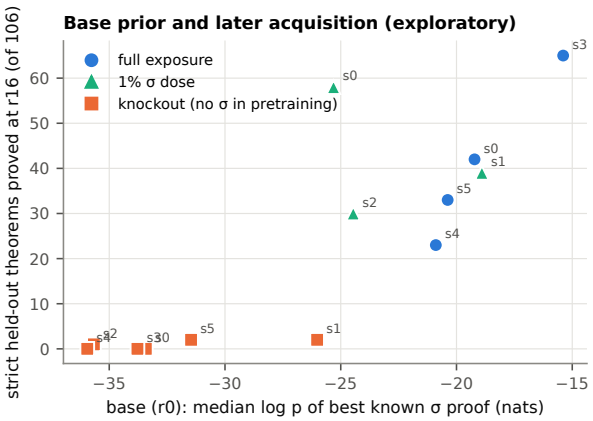


Figure 6: Exploratory (*this paper*): the base's median log-probability of known σ proofs against strict held-out theorems proved at r16, for the 13 ladders with a scored base.

and so is `916`. The fourth member, `1019`, had no base success in about 10^4 attempts, and it is the one that gained most. Conversely, knockout seed 4's base proved three converse-contraposition members at 1.2–4.9 $\times 10^{-4}$, and its ladder proves no strict theorem at r16. A base rate near 10^{-4} thus went with amplification on at most one of the four seeds whose bases had such members. A re-sample of seed 1's members at $n \geq K$, run for this paper, is pending; at that sample size it can classify only members whose rate is well above or below $1/K$.

3.5 What else the knockout changes

Data. Removing σ^+ records also thins its constituents. In $\mathcal{D}^-$, `DN` appears in 3.29% of records (13.73% in $\mathcal{D}$) and $\neg$-introduction in 5.19% (15.21%). Theorems with no intuitionistic proof fall from 4.17% to 1.35% of a 20,000-record sample; the rest use `DN` on a $\neg\neg$ that a premise or an assumption supplies. Mean proof term size moves from 35.91 to 36.25 nodes. The knockout removes the composition, and it also makes each constituent three to four times rarer.

General ability. The knockout bases are 6.88 points worse at greedy decoding of 5,000 held-out generator theorems (0.8829 vs 0.9517). The gap comes from the 679 theorems whose reference proof uses σ (0.45 vs 0.96); on the other 4,321 the knockout is 0.12 points better. With 256 attempts, though, the knockout bases also prove 4.3 fewer of the 225 intuitionistic theorems of holdout250, and 4.0 fewer of textbook72's 53 (three full seeds against six knockout seeds). By r16 the holdout250 gap closes (0.996 vs 0.996), while textbook72 keeps −2.5.

Gains elsewhere. On the 322 evaluation theorems, the base was given 512 attempts and then at least $K_{\text{train}}(16)$ on every theorem it failed. Within that budget the knockout bases reach 241–259 theorems and the full bases 255–261; at r16 the knockout models prove 277–286 and the full models 291–298. RL's gain over what its own base reaches (Δ_{cov}) is +37, +36 and +32 for full seeds s3–s5 and +21 to +37 for the knockout seeds. The means are 35.0 and 29.8, one-sided Mann–Whitney $p \approx 0.13$ with one tie (*this paper*). Part of the knockout's gain is catching up on its own pretraining deficit. The knockout ladders may be somewhat worse at RL in general. They are not clearly

so, and the difference is small beside 8–69 against 0–2 strict theorems. These sets are not netted against RL-free controls.

What this means for the interpretation. Two things favour exposure to the composition over a general deficit. The general costs are a few theorems out of hundreds. And the dose arm restores acquisition while moving `DN` and ¬-introduction by about one point each (§3.2). The design cannot separate exposure in pretraining from exposure through replay during RL (§2). Nor can it separate exposure to the composition from the constituents' frequency, which a matched-frequency knockout would [1, 22].

3.6 The base's prior predicts what is proved at r16 (exploratory)

Across the 13 ladders with a scored base, the base's median log-probability of known σ proofs is rank-correlated with strict held-out theorems proved at r16 (Spearman $\rho = 0.88$; Fig. 6). This is mostly the difference between arms: $\rho = 0.39$ ($p = 0.38$) without the knockout. Within a ladder, the base's log-probability of a member's best known proof predicts whether the member is proved at r16. The AUC is 0.92–0.95 on the four full seeds with scored bases (pre-registered expectation ≥ 0.7) and 0.85–0.86 on the three dose seeds. Restricted to members not proved at r0, so that only RL-phase gains count, it is 0.83–0.93 (full) and 0.74–0.84 (dose). Comparing only members of the same family, the AUC is 0.47–0.98 (full) and 0.28–0.60 (dose). One full seed is higher within families than overall, against the expectation stated before computing. The base's log-probability also predicts *when* a member is first proved in the ladder's per-round samples. The Spearman ρ with the first-solve round is -0.69 to -0.87 on the full seeds and -0.61 to -0.72 on the dose seeds. Within families it is -0.33 to -0.48 (full) and -0.22 to $+0.14$ (dose). On the dose seeds, then, the prior predicts across families and not within them (all *this paper*). Reference-solution likelihood predicts success at larger k [39], and RL gains concentrate where the base already succeeds sometimes [61].

3.7 Two nulls

One seeded round. From each knockout r16 checkpoint we ran one more EI round, with and without four copies of one free positive reductio proof written by knockout seed 1's base (of `la_transfer_916`). Seeded minus control was +0, +0, +2, +0 and +0 strict family members on seeds 0, 1, 2, 3 and 5 (seed 4 was not run). Seed 2's two were single successes in 256 attempts (one-sided exact $p = 0.25$). This contrasts with the earlier, smaller organism, where a single injected sibling proof put the reductio on 6 and 17 targets in the next round (§6). Two explanations remain open: this organism resists one-proof seeding, or the read-outs differ. The injected proof here is of a non-family theorem, and the outcome is strict held-out members rather than training targets.

Power sampling. Metropolis–Hastings sampling from p^α of the full and knockout bases [21] ($\alpha = 2, 4$; 32 chains per theorem) proved none of the theorems RL r8 proves beyond the base's reach (0/33, 0/27), and none of the 106 strict theorems for knockout bases s0 and s1. Plain sampling with the same decoded tokens proved 1–2. The sampler also failed its positive control: on 17 theorems the base proves rarely, it proved 0/17, where plain sampling at the same tokens proved 6/17 ($T = 0.8$) and 13/17 ($T = 1.0$). This is a null for this sampler at these settings, not for sharpening. Practical power samplers do not reach p^α at realistic budgets [45]. Details are in Appendix F.

3.8 Robustness

The independent re-check accepted every stored knockout proof of a strict theorem in Lean 4.34.1 (448/448), the version used in the loop. It rejected all 150 re-submitted Lean rejections, all 60 proofs paired with the wrong theorem, all 40 strict knockout proofs with `Classical.byContradiction` removed, and `sorry`. Every rejection in the experiment was the pre-filter's, so its false-reject rate matters. Re-checking 150 rejections per reason for its 13 reasons gave 0 accepted, a 95% bound of 2.0% per reason. This paper Lean-checked every pre-filter rejection in knockout seed 1's ladder (222,135 texts, rounds 1–16; 22,522 on a by-contradiction step); the result is pending. That check covers the ladder's samples, not the large-k samples behind the certificates, whose rejected texts were not stored. Cut-offs (a step over 512 tokens or a proof over 96 steps) can only hide successes. On the strict theorems at r16 they affect 0.04–4.97% of knockout attempts, 0.11–2.89% of full and 0.68–8.84% of dose attempts. A paired re-evaluation with doubled caps, run for this paper, has so far covered 4 of 10 checkpoints, with counts moving by at most one theorem (Appendix I). The training theorems share no renaming class with any evaluation set, and no pretraining set shares one with the 106 strict theorems.

4 How much did RL create in the full-exposure models?

The knockout asks whether EI can acquire a composition the base was not shown. Most claims in the literature are instead about a single RL run and its own base. Before the knockout, the project applied the protocol of §2 to the three earlier full-exposure seeds (s0–s2, r0 vs r8, on the 322 evaluation theorems). This section summarises what it found. Every number is from reviewed reports of those runs (Appendix J).

The budget. The five allocations of §2 give five values of K (Table 7). K_{pub} is the value an earlier report used: the ladder's RTX A6000 seconds divided by a per-attempt cost measured on hard theorems on an A40. The base costs 6.36–8.04 ms per attempt on these theorems on an A40, and we convert A6000 seconds at a measured A6000/A40 ratio of 0.932. The compute-matched set is a curve in K (Fig. 7). Over a 12–16-fold range of K it falls from 34/33/33 to 21/20/14 theorems for s0/s1/s2. These 21/20/14 are compute-matched under all five allocations, and 18/17/8 of them had no base success in at least 66,304 attempts. At the level of sets, RL stays ahead of its base under every allocation and seed: Δ_{cov} is $+21/+16/+12$ at K_{pub}. Unlike the large-k crossovers reported for LLMs [59], the base here does not catch up at 10^4–5×10^4 attempts per theorem.

Brackets, certificates and controls. Sampling resolves success rates only down to about $1/n$. Below that we estimate them from likelihoods. The *known-proof estimate* sums the base's probability, teacher-forced at $T = 0.8$ over the 33 line-numbering offsets, of every Lean-accepted proof of the theorem that any model found. It recovers a median 0.98/0.94/0.93 of the measured rate on theorems with 3–100 successes in 512 attempts. It runs about two-fold low at 10^{-5}–10^{-4} (median ratio 0.45) and lower still below (0.36 on two theorems; Appendix D). Because the ratios are binned by measured rate, they condition on at least one success. Of the 9 pre-selected pairs, 7 certify at K_{pub}. Four of them, on three theorems, also had no success from the RL-free controls (512 attempts each) or from any other seed's base; only one of the three, a Peirce instance, is classical-only. The two controls receive RL's training without RL's proofs:

- *replay-only*: eight fine-tuning rounds on pretraining replay alone. It proves 239/244/249 of the 322, against the base's 232/236/234 and r8's 286/286/293.

Table 7: What survives each filter, full-exposure seeds s0/s1/s2 (r8 against its own base). The sets shrink by a factor of 2.6–4.3 from equal k to the strictest budget. The rows below $\mathcal{C}(K_{\text{adapt}})$ are separate filters, not a nested funnel: two start from $\mathcal{C}(K_{\text{pub}})$, and the last two come from 9 (theorem, seed) pairs pre-selected as the most reliable r8 proofs in each seed's strictest set and sampled 1.1–1.5 million times each.

filter	budget K (s0/s1/s2)	theorems
RL proves, base 0/256 (equal k)	256	54/51/60
$\mathcal{C}(K_{\text{att}})$: ladder sampling attempts on training theorems	3,574	34/33/33
$\mathcal{C}(K_{\text{train}})$: sampling + fine-tuning GPU time	7,416/7,126/9,921	33/27/24
$\mathcal{C}(K_{\text{uni}})$: all ladder GPU time, same GPU class	10,314/10,141/14,150	28/26/24
$\mathcal{C}(K_{\text{pub}})$: earlier report's value (mixed GPU classes)	19,281/21,107/23,310	22/21/14
$\mathcal{C}(K_{\text{adapt}})$: 512 each, the rest on the base's failures	44,570/43,427/56,509	21/20/14
of which no base success in $\geq$ 66,304 attempts		18/17/8
$\mathcal{C}(K_{\text{pub}})$, net of the replay-only control		20/14/8
$\mathcal{C}(K_{\text{pub}})$, never proved by any seed's base		4/5/5
9 pre-selected pairs: certified out of reach at K_{pub}		7 of 9 (0 of 9 at K_{adapt})
and no success from any RL-free control or other seed's base		4 pairs on 3 theorems

- *continued pretraining*: about 75–78% of the ladder's GPU time. It proves 220/215/219, gaining 12–19 theorems and losing 30–42.

The cheaper control is informative: replay alone recovers 13–25% of RL's gain on the 322.

Likelihoods along RL. On theorems RL newly proves, the hardest step of r8's most likely proof rises from about -6.2 nats at r0 to about -1.3 at r8, mostly by round 4 (teacher-forced at $T = 1$). Part of this is by construction, since the proof is selected at r8. RL also lifts the pretraining reference proofs of the same theorems, about three times less. On theorems neither r0 nor r8 proves, the reference proofs do not improve.

Test-time compute before and after RL. If RL mostly converts reachable-but-unlikely proofs into likely ones, extra compute at test time should help the base more than the RL model (Fig. 8). A guided evaluation adds 18–22 holdout250 theorems to the base and -1 to $+4$ to the RL models. On holdout250 the RL models are within 3–14 theorems of the ceiling, and guided reading halves the gap between the base and r8 (44/31/37 theorems plain, 21/14/19 guided). Textbook72 is the fairer pool: there, guided redraws add 8–12 to the base and 3–6 at r8. Guided attempts cost about 2.2 times the GPU time of plain ones, so this is not compute-matched. At matched sampled tokens, guided redraws still beat plain resampling on long textbook problems for six r8 models, by 5.2–10.3 points. A PUCT tree search over proof states, with a value head trained on the frozen policy's rollouts, was compared with sampling at equal wall-clock. At r8 it is within ± 3 theorems of sampling on every pool, and it failed its pre-registered gate on the theorems neither r0 nor r8 proves. At r0 it adds 10–19 theorems on 100 long theorems (13–16 lines), about 44% of what eight EI rounds add there by sampling. That fraction is 14% on holdout250 and 17% on textbook72. Neither method reaches most of the compute-matched creations: the guided base proves 3/4/1 of the 22/21/14 at K_{pub}. DeepSeek-Prover-V1.5 reported its search and RL gains as orthogonal [53]. Here they overlap: search recovers part of RL's gain at the base and adds little after RL.

5 Related work

Does RL expand what the base can do? Equal-k comparisons in LLMs find base models overtaking RL as k grows [59]. Others report gains where the base fails, read at $k = 128$ [28].

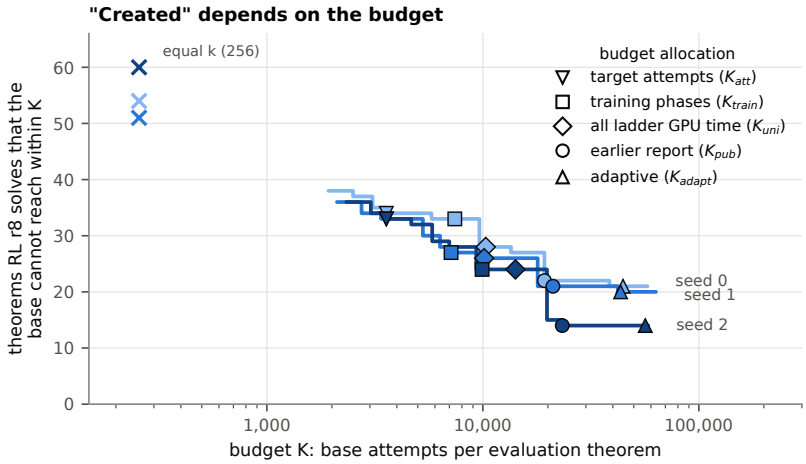


Figure 7: The compute-matched set (RL r8 proves, its base cannot reach within K) against the base budget K, for the three earlier full-exposure seeds; markers are the five allocations of Table 7; crosses mark equal k ($K = 256$; 54/51/60). Lines stop at 66,304 attempts, the most any theorem received; beyond it, membership cannot be decided.

Early RL can shrink the boundary of what is solved while prolonged RL expands it [57], and wider rollouts revive plateaued runs [16]. Much of RL's measured gain corresponds to operating points the base reaches with search at small sample counts [44]. In five public RLVR pairs, the visible pass@k crossovers were not statistically established [55], and reward and contamination controls change conclusions [40, 50]. These studies compare base and RL at equal k, or through exchange rates up to 16 samples [44]. None ties the base's budget to RL's training compute, which we do, reporting the created set as a curve.

Controlled organisms and pretraining exposure. The closest design is Abdulsalam et al. [1]. They pretrain on primitive rewrites that exclude later shortcuts, read the base at pass@1024 and RL at pass@16, and find GRPO building the shortcuts. Rejection-sampling fine-tuning, the family our EI belongs to, "then plateaus, while RL improves more slowly at first but continues climbing". Pretraining at a low contraction weight ρ "never reliably enters the macro or parallel regime". Their RFT and low-ρ results resemble ours; their GRPO result is the contrast. Tsilivis et al. [46] set the share of long demonstrations exactly and show RL amplifying that mode efficiently unless the share is exponentially small. Zhang et al. [61] find no RL transfer to a context with 0% or 0.1% pretraining exposure and transfer at 1%, with GRPO in a 100M model; our dose result parallels theirs for a composition of two rules. Pretraining with more reasoning data leaves a lead that survives RL at 8B [2]. Priming with demonstrations of a behaviour enables RL to use it [11]. RL composes new strategies only from atomic skills already mastered [7]. If σ's two rules count as mastered here, the knockout is a counterexample for EI at this scale; if not, rules at a quarter to a third of their former frequency are not mastered. RL composes known string functions where iterated RFT "never surpasses 2.6%" [58]. Knockouts after pretraining disagree. With a strategy removed at the fine-tuning stage, "we do not observe novel patterns emerging through reinforcement learning" [24]. With a pattern family removed from the RL data, a 1.5B model "successfully learns to generate the held-out patterns despite zero training exposure" [31]. Neither measures the removed behaviour's base rate. In from-scratch organisms, RL collapses onto one pretraining mode [62] and surfaces chess moves to which fine-tuning gave probability below 0.05

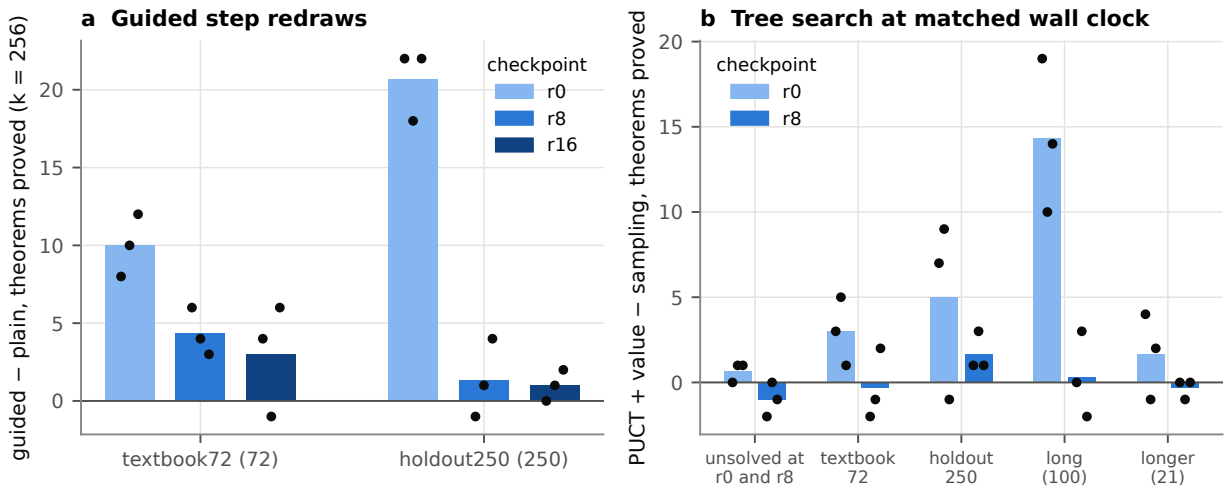


Figure 8: Test-time compute helps the base more than the RL model. (a) Guided minus plain evaluations ($k = 256$ each) for the base (r0) and after 8 and 16 rounds; bars are means, dots are seeds. Not compute-matched: guided attempts cost about 2.2 times the GPU time. (b) PUCT search with a learned value head minus sampling at equal wall-clock. "Group C" holds the theorems neither r0 nor r8 proves in an earlier evaluation (28–36 per seed); "long rrQ100" holds 100 theorems of 13–16 lines; "long2" holds 21 longer theorems.

[41]. Small from-scratch provers and solvers also improve on their own filtered outputs [26, 32]. On graph path-finding, 0/1-reward policy gradient without a KL term equals fine-tuning on the model's own successes, yet "can explore and discover new correct paths that were absent from the initial training set" [48]. Our knockout keeps both constituents of the removed composition, refills the data, and measures the base's rate for the composition at about a million attempts.

Why on-policy RL may stay near the base. With an outcome reward, the number of reward queries policy gradient needs grows as the base's likelihood of a solution shrinks, and going beyond the base's support can cost exponentially much [29]. "Passive exploration" is unlikely to "generate novel and correct proofs by chance" [52]. On-policy RL prefers solutions close to the base [42], and the KL-regularised optimum with a binary reward re-weights the base toward success [23]. On hard problems, on-policy RL rarely sees a correct rollout [36], and in a one-layer theory, outcome-reward policy gradient "becomes infeasible" when the data's mass on simple examples vanishes [37]. Negative samples matter too: training only on the model's failures keeps the base's pass@256, while training only on successes lowers it [63]. In GRPO on a Lean prover, gains fade at large N because rare correct proofs are seldom uplifted [14]. In LLM reasoning, EI matched or beat PPO, and RL models "fail to explore significantly beyond solutions already produced by SFT models" [13]. Our knockout measures this limit directly: the base's rate for the missing composition, and whether EI amplifies it.

Sharpening without training. Metropolis–Hastings sampling from the base's power distribution p^α approaches GRPO on MATH500 at 8.84 times the tokens [21]; a token-level approximation is reported to match one-shot GRPO [19]. Practical power samplers, however, do not converge to p^α at realistic budgets [45], and one α sharpens different problems very differently [56]. Our sampler failed its own positive control, so our null concerns the sampler.

Elicitation and measurement. A latent capability is one a model shows rarely by default and that small interventions reveal [15, 47]. Password-locked models show that a few demonstrations, or expert iteration from lucky samples, recover hidden skills, while RL from a weak policy can fail to explore [12]. Fine-tuning often enhances or wraps existing mechanisms [18, 35]. Li and Liu [27] define creation as a change in the support a model reaches under a stated budget, decoding policy and signal, and a control that never had the capability makes such claims checkable [9]. With unlimited samples, any model with full support covers everything [5]. Rare events need importance sampling, extrapolation or sampling until a fixed number of successes [17, 20, 49], and reference-solution likelihoods predict success at larger k [39]. Thresholded metrics can manufacture jumps [38], so we report our family criterion beside counts. In compositional generalisation, compounds can be held out while atoms stay [22], and composing rare concepts "requires considerably more optimization steps" [30]. Our knockout holds out a compound, but it thins the atoms too (§3.5).

Expert iteration for theorem proving. Neural provers run the sample–verify–train loop with search as the expert [25, 33, 34, 51, 54] or with GRPO on a Lean reward [53]. The loop stalls when hard targets yield no successes: a from-scratch HOL prover stays at 7% without injected exploration [4], and a self-play prover found 131 correct proofs in 2.5 million attempts on unproved statements [10]. At least one formal-mathematics creation claim rests on a 64-attempt base budget [6].

6 Discussion

What the knockout shows. Without σ in pretraining or replay, 16 rounds of this EI acquired no classical family on any of six seeds. With it, full-exposure seeds gained 8–69 strict held-out theorems during the RL phase, and a 1% dose was enough on three seeds. The pre-registered verdict, NEEDS EXPOSURE, holds for converse contraposition, the family the full bases mostly had before RL; for the others it is undetermined at six seeds. Beyond the waiting-time null (post hoc), the knockout ladders raised their own rate of the free positive reductio: three reached a strict training theorem sooner than their base rates predict. They did not raise it to a family.

Four readings of the null.

1. *σ is out of the policy's reach.* This is false for five of six bases, which proved strict theorems with σ at large k, and for four of six under guided decoding at $k = 256$. Seed 0's base is certified out of reach only for plain sampling, at one budget and with an 8% margin; its guided evaluation proved a strict theorem.
2. *EI samples too little.* At 32 attempts per theorem, EI's per-round budget on the strict theorems is about 4,000 attempts, against base rates near 10^{-6}. That explains why most ladders saw nothing for many rounds. It does not explain everything: three ladders reached strict theorems sooner than this null predicts ($p = 0.006$–0.019).
3. *EI amplifies rare finds too slowly, or too narrowly.* Seed 5 wrote a free positive reductio in round 2 and did not spread it in 14 more rounds. Seed 1 multiplied its free positive reductio more than tenfold in eight extra rounds and still proved the same five held-out theorems. In EI a single new proof enters a fine-tuning mix of about 70,000 RL records (up to four proofs for each of 4,495 theorems, each repeated four times) plus 20,000 replay records.
4. *The knockout made the bases worse learners in general.* The general costs are a few theorems out of hundreds, and the knockout's gains elsewhere are not clearly smaller (partly catch-up). The

dose arm restores acquisition while moving the constituents by about one point. This reading is not excluded, but it is not needed.

Positive-only or signed updates? Whether another RL algorithm would do better from the knockout bases is the main open question. EI updates only on successes. GRPO also pushes down failures and gives a lone success in a group of 32 an advantage of $\sqrt{31} \approx 5.6$ standard deviations. That helps only after a first success: when all 32 attempts fail, the group baseline gives no gradient, so on the strict theorems GRPO at $k = 32$ faces the same waiting time as EI. The evidence cuts both ways.

- **Against EI-specificity.** In this organism, from the same full-exposure bases at equal sampled attempts, GRPO did not prove more theorems than EI. Default and unlikeliness-weighted GRPO proved significantly fewer of the 322 (theorems solved only by GRPO vs only by EI: 15 vs 38, sign test $p = 0.0022$; 15 vs 44, $p = 0.0002$). A pass@4 advantage was level (33 vs 20, $p = 0.098$), ahead on the hardest theorems mostly through one seed. Interplay's 0% exposure null used GRPO [61]. A strategy removed at fine-tuning did not re-emerge under GRPO [24]. In LLM reasoning EI matched or beat PPO [13].
- **For EI-specificity.** Rejection-sampling fine-tuning plateaued where GRPO built new shortcuts [1], iterated RFT failed where RL composed functions [58], and GRPO produced a pattern family absent from its RL data [31]. Training on negative samples preserves breadth that positive-only training loses [63]. In our earlier 3.2M organism, GRPO acquired a depth-3 nesting habit from zero-rate bases where EI mostly did not (below).

GRPO has not been run on the reductio in any of our organisms. A GRPO arm from the six knockout bases, with per-seed predictions written down first, is the decisive experiment. By sampling alone, only the seeds with a strict success inside 16 rounds (1, 2, 5) could be rescued, so any further rescue would have to come from drift.

Earlier organisms. The project studied related patterns in smaller organisms: 3.2M–85M-parameter models that write whole proofs, pretrained on proofs of at most six lines (their RL proofs ran to seven or eight). They were judged by an older in-house checker, with Lean confirmation where stated. Their criterion for *igniting* a pattern was at least 2% of the targets that need it, much weaker than our family bar.

- **Reductio, required pools.** Four of eleven reductio-free bases emitted the strict reductio before RL (8×10^{-6}–2.3×10^{-4} per sample), and exactly those four ignited under EI. First proofs arrived about when $1/(rkN)$ predicts (rounds 1, 2, 6 and 6, against 0.2, 1.7, 2.6 and 6.2). The other seven stayed at 0 of 606. In those pools every target needed the pattern, so a zero-rate base gave EI no training step at all. At 3.2M, 25M and 85M parameters the rates did not rise with size. A rare 85M base that read 0 of 300 after eight rounds produced its first proof in round 13 when continued. Igniting arms saturated the seven-line targets and went no further, an analogue of knockout seed 1's narrow spread.
- **Injected data.** One fine-tuning step on a sibling's found proofs ignited all 11 dead arms. A single injected proof ignited the reductio on two dead seeds, reaching 6 and 17 targets in the next round. Four unrelated generator instances, and even four invalid pattern strings, ignited some arms. In the present organism, 16 demonstrations taught excluded middle (35–37 of 39 members) to a base pretrained with no `DN`.
- **Mixed pools.** Where EI could train on other proofs, it ignited a depth-3 nesting habit from zero on three of five bases, decided during training rather than by sampling luck. GRPO ignited it on every surviving arm from two zero-rate bases (12 of 12), where paired EI did so in 1 of 3.

That habit was not needed for the targets, the zero rates were measured with the older checker, and the checkpoints are lost.

The knockout fits these results only partly. As there, the base rate predicts which seeds move, and no ladder that started at a pooled rate near 10^{-6} reached the family bar. Unlike the required pools, our mixed pool let the knockout ladders' rate rise: by the earlier criterion, knockout seed 1 ignited at round 13, and three ladders reached strict theorems sooner than their base rates predict. "Needs exposure" should therefore be read as "needs exposure, demonstrations, or perhaps a different update, for this EI to acquire the family within 16 rounds".

An amplification threshold? One might expect EI to amplify a composition once some training theorems' rates are high enough for 32 attempts per round to hit them within a few rounds. Our data are mixed. Knockout seed 1 amplified members its base proved at about 10^{-4}, but its largest gain was on a member with no base success. Knockout seed 4's base proved three members at 1.2–4.9×10^{-4}, and its ladder amplified none. Two doses cannot tell a threshold from a graded response. The median likelihood of known proofs also misses it: the nearest dose and knockout bases are within 0.7 nats, yet their outcomes differ by 28–56 theorems. The 0.1% dose, and per-theorem rates on training theorems at several checkpoints, would test it.

What transfers to larger models. The protocol transfers: budgets matched to RL's compute and reported as curves, RL-free controls, certificates. So does the waiting-time logic. A first success is expected after about $1/(rkN)$ rounds, so with 10^4–10^5 prompts and k of 8–64, LLM RL can find far rarer compositions than our 123 strict theorems allow, if a comparable share of prompts needs them. The numbers do not transfer. Two practical points follow, with that caveat.

- **Evaluation.** Absence at modest k under plain sampling is not absence. Our knockout bases fail at $k = 256$, yet a million plain attempts or a guided evaluation at $k = 256$ reveals the composition.
- **Data filtering.** Removing a composition from pretraining kept this EI from acquiring it for 16 rounds, but 1% of records restored it. We did not test the leakage levels a real filter leaves. In the same organism, 16 demonstrations taught a no-`DN` base excluded middle, so fine-tuning on demonstrations would likely defeat such a filter.

An evidence hierarchy for "RL created it". In increasing strength:

1. success at equal k;
2. a compute-matched base budget, reported as a curve over defensible allocations;
3. RL-free controls with matched training;
4. a certificate against the base at that budget;
5. an intervention on what the base saw.

For our full-exposure seeds the claim shrinks at each step (Table 7). With the intervention, exposure was followed by acquisition within 16 rounds, and its absence by none, for one composition and one algorithm. Steps 2 and 3 cost about as much as the RL run they audit, so they are affordable wherever that run was; they are rarely reported.

7 Limitations

- **Scale and domain.** One architecture with 9.56M parameters, propositional logic over at most four atoms, and pretraining proofs of at most 12 lines. Nothing here is measured on a large

language model or on first-order logic. The organism was chosen because it allows interventions that large models do not, and its results are results about it.

- **One RL algorithm, 16 rounds, one composition.** Expert iteration at $k = 32$, $T = 0.8$ is rejection-sampling fine-tuning. GRPO-style updates, more exploration, demonstrations or longer runs could behave differently, and in the closest controlled study GRPO did, and in our earlier organism GRPO acquired a depth-3 nesting habit faster than EI; in this organism, from full-exposure bases, plain GRPO proved fewer theorems than EI (§6). One knockout ladder was still changing at round 24. Only one composition was removed.
- **What the headline test measures.** The pre-registered indicator is largely decided before RL. The RL-phase contrasts (gains during the RL phase, families acquired during it) were not pre-registered, and the RL phase includes replay. On the three full seeds that match the knockout's hardware, the gains test gives $p = 0.012$ and the family contrast 2/3 vs 0/6 ($p = 0.034$).
- **Seeds and power.** Six seeds per main arm power the primary indicator but not most per-family tests (three families 3 vs 0, Holm 0.17; excluded middle 1 vs 0). A 0/6 outcome bounds the per-seed rate of acquisition after σ-free pretraining below 0.39 only. The dose arm has three seeds and two doses (0% and 1%) cannot locate a threshold.
- **The knockout is not surgical, and exposure includes replay.** It makes `DN` and ¬-introduction three to four times rarer and removes two thirds of the classical-only theorems. It costs a few theorems of general ability (§3.5). Exposure in pretraining cannot be separated from exposure through replay during RL.
- **"Absent" is not established.** Five of six knockout bases proved strict theorems with σ at large k, and four of six under guided decoding at $k = 256$. Only one is certified out of reach as pre-registered: under plain sampling, on a pooled mean rate, at one allocation, with an 8% margin. Spreading the same compute over 106 or 1,054 theorems would certify 0 or 5 of 6. The certificate assumes the pre-filter rejected no valid proof in the large-k sample, which could not be checked. No family could be certified at the sampling used, so the CREATED verdict was unreachable by design. The "rare" evidence comes almost entirely from one family (converse contraposition); on the others the bases sit at the sampling floor.
- **Within-reach labels.** Per-theorem base rates in the knockout arm rest on one to three successes in fewer than K attempts, from sampling that paused after a first success.
- **Estimators.** The known-proof estimate runs two- to three-fold low below 10^{-4} and was calibrated on full-exposure bases. The likelihood scores use the full ladders' proofs and one line-numbering offset.
- **Budgets.** K depends on the allocation, on the number of evaluation theorems it is spread over, and on one measured GPU-class ratio. The knockout certificates and Δ_{cov} were computed at one allocation.
- **Settings.** Old and new full seeds differ in GPU class, batch, resumption and evaluation-phase sampling, and are pooled. Many evaluations cut off more than 0.1% of attempts in some stratum (up to 8.8% of strict-theorem attempts on one dose seed); see Appendix I. The power-sampling comparators kept no proof text, so their few successes were not re-checked in Lean.
- **Interface.** In an earlier 3.2M organism, a base acting through the proof-state interface reached most of the theorems that whole-proof EI had "created", with two seeds per arm and the state confounded with step-by-step generation. Whether creation verdicts depend on the interface is untested at this scale.

8 Reproducibility

Code, data and models. All code is in the fork `dan-pandori/nd-takehome`. The experiment is on branch `dan_create-or-elicit` (commit `8e195390`, with its review): scripts `coe/` and `pod/coe/`, the independent recount code `review_coe/`, the pre-registration, per-round statistics, budgets, the estimator calibration and a claims ledger. This paper is on `dan_create-or-elicit-paper`. Its figures and statistics are regenerated by `paper/scripts/` from those files, and every number in the text has a row in `paper/LEDGER.md` that points to its source. The pretraining sets, all 200 ladder checkpoints, the evaluation rows, the large-k samples and the Lean-gate dumps are in the public bucket `hf://buckets/dan-pandori/nd-rl/create-or-elicit/`. The decompressed md5 sums of the pretraining sets are $\mathcal{D}$ `800b5486`, $\mathcal{D}^-$ `466fc214` and $\mathcal{D}^{1\%}$ `19cd2263`.

Pre-registration and review. The experiment's pre-registration was committed before its first GPU pod. Six dated amendments each precede the work they govern, except the strict-set amendment (§3.1), which came after round 1 of the first knockout ladder and before any acquisition outcome. The independent recount re-derived the experiment's counts in §3 from the raw files; numbers marked *this paper* are computed here from those files. The experiment's pre-registration was written after the earlier full seeds' ladders had run, so their training-side dynamics were known when it was written. It corrected three Boschloo p-values that had been computed on a transposed table; no verdict changed (Appendix H). This paper's own checks and analyses were pre-registered as well (`preregistration/create-or-elicit-paper.md`, Appendix I). Two of them are post hoc, and labelled so. The experiment's pre-registered expectations are compared with outcomes in Appendix G.

Compute. Table 8 gives the compute per arm. Every 16-round ladder made 2,301,440 attempts on training theorems and 9,600 fine-tuning steps, and the knockout and dose ladders' compute is within 0.95–1.03 times the full ladders' on every quantity. The experiment used 16 GPU instances (15 NVIDIA A40 at \$0.59/h and one RTX A6000) for 309.34 instance-hours and \$182.50. This paper's checks used two A40 instances (final hours and cost: pending).

Table 8: Compute per new 16-round ladder, mean over seeds (A40; results-registry rows, recounted independently). Pretraining took a further ≈1,280 GPU-seconds, ≈24,000 steps and $\approx 4.4 \times 10^8$ tokens per seed.

quantity	KNOCKOUT	FULL (s3–s5)	DOSE
sampling GPU-s	32,441	34,005	34,492
evaluation GPU-s	7,886	8,046	7,970
fine-tuning GPU-s	13,747	13,753	13,898
total GPU-s	54,105	55,841	56,390
generated tokens	6.69×10^8	6.73×10^8	6.76×10^8
Lean checks	2.15×10^6	2.21×10^6	2.18×10^6

Acknowledgements

Left for Dan.

References

[1] Azwar Abdulsalam, Nishil Patel, and Andrew Saxe. RL Post-Training Builds Compositional Reasoning Strategies. arXiv:2607.07646, 2026.

[2] Syeda Nahida Akter, Shrimai Prabhumoye, Eric Nyberg, Mostofa Patwary, Mohammad Shoeybi, Yejin Choi, and Bryan Catanzaro. Front-Loading Reasoning: The Synergy between Pretraining and Post-Training Data. arXiv:2510.03264, 2025.

[3] Thomas Anthony, Zheng Tian, and David Barber. Thinking Fast and Slow with Deep Learning and Tree Search. arXiv:1705.08439, 2017.

[4] Kshitij Bansal, Christian Szegedy, Markus N. Rabe, Sarah M. Loos, and Viktor Toman. Learning to Reason in Large Theories without Imitation. arXiv:1905.10501, 2019.

[5] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large Language Monkeys: Scaling Inference Compute with Repeated Sampling. arXiv:2407.21787, 2024.

[6] Zachary Burton. Inference-Time Diversity in RL-Trained Lean Theorem Provers: A Diagnostic Study. arXiv:2601.16172, 2026.

[7] Sitao Cheng, Xunjian Yin, Ruiwen Zhou, Yuxuan Li, Xinyi Wang, Liangming Pan, William Yang Wang, and Victor Zhong. Atomic Skills are the Prerequisite: When Reinforcement Learning Synthesizes Compositional Reasoning, and When It Only Amplifies. arXiv:2512.01970, 2025.

[8] Tom Davidson, Jean-Stanislas Denain, Pablo Villalobos, and Guillem Bas. AI capabilities can be significantly improved without expensive retraining. arXiv:2312.07413, 2023.

[9] Aghyad Deeb and Fabien Roger. Do Unlearning Methods Remove Information from Language Model Weights? arXiv:2410.08827, 2024.

[10] Kefan Dong and Tengyu Ma. STP: Self-play LLM Theorem Provers with Iterative Conjecturing and Proving. arXiv:2502.00212, 2025.

[11] Kanishk Gandhi, Ayush Chakravarthy, Anikait Singh, Nathan Lile, and Noah D. Goodman. Cognitive Behaviors that Enable Self-Improving Reasoners, or, Four Habits of Highly Effective STaRs. arXiv:2503.01307, 2025.

[12] Ryan Greenblatt, Fabien Roger, Dmitrii Krasheninnikov, and David Krueger. Stress-Testing Capability Elicitation With Password-Locked Models. arXiv:2405.19550, 2024.

[13] Alex Havrilla, Yuqing Du, Sharath Chandra Raparthy, Christoforos Nalmpantis, Jane Dwivedi-Yu, Maksym Zhuravinskyi, Eric Hambro, Sainbayar Sukhbaatar, and Roberta Raileanu. Teaching large language models to reason with reinforcement learning, 2024.

[14] Andre He, Daniel Fried, and Sean Welleck. Rewarding the Unlikely: Lifting GRPO Beyond Distribution Sharpening. arXiv:2506.02355, 2025.

[15] Felix Hofstätter, Teun van der Weij, Jayden Teoh, Rada Djoneva, Henning Bartsch, and Francis Rhys Ward. The Elicitation Game: Evaluating Capability Elicitation Techniques. In *International Conference on Machine Learning (ICML)*, 2025.

[16] Jian Hu, Mingjie Liu, Ximing Lu, Fang Wu, Zaid Harchaoui, Shizhe Diao, Yejin Choi, Pavlo Molchanov, Jun Yang, Jan Kautz, and Yi Dong. BroRL: Scaling Reinforcement Learning via Broadened Exploration. arXiv:2510.01180, 2025.

[17] Shengding Hu, Xin Liu, Xu Han, Xinrong Zhang, Chaoqun He, Weilin Zhao, Yankai Lin, Ning Ding, Zebin Ou, Guoyang Zeng, Zhiyuan Liu, and Maosong Sun. Predicting Emergent Abilities with Infinite Resolution Evaluation. arXiv:2310.03262, 2023.

[18] Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, Robert P. Dick, Hidenori Tanaka, Edward Grefenstette, Tim Rocktäschel, and David Krueger. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. arXiv:2311.12786, 2023.

[19] Xiaotong Ji, Rasul Tutunov, Matthieu Zimmer, and Haitham Bou Ammar. Scalable Power Sampling: Unlocking Efficient, Training-Free Reasoning for LLMs via Distribution Sharpening. arXiv:2601.21590, 2026.

[20] Erik Jones, Meg Tong, Jesse Mu, Mohammed Mahfoud, Jan Leike, Roger Grosse, Jared Kaplan, William Fithian, Ethan Perez, and Mrinank Sharma. Forecasting Rare Language Model Behaviors. arXiv:2502.16797, 2025.

[21] Aayush Karan and Yilun Du. Reasoning with Sampling: Your Base Model is Smarter Than You Think. arXiv:2510.14901, 2025.

[22] Daniel Keysers, Nathanael Schärli, Nathan Scales, Hylke Buisman, Daniel Furrer, Sergii Kashubin, Nikola Momchev, Danila Sinopalnikov, Lukasz Stafiniak, Tibor Tihon, Dmitry Tsarkov, Xiao Wang, Marc van Zee, and Olivier Bousquet. Measuring compositional generalization: A comprehensive method on realistic data. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:1912.09713.

[23] Tomasz Korbak, Ethan Perez, and Christopher L. Buckley. RL with KL penalties is better viewed as Bayesian inference. In *Findings of EMNLP 2022*, 2022.

[24] Akshay Krishnamurthy, Audrey Huang, and Nived Rajaraman. Select and Improve: Understanding the Mechanics of Post-Training for Reasoning. arXiv:2606.13125, 2026.

[25] Guillaume Lample, Marie-Anne Lachaux, Thibaut Lavril, Xavier Martinet, Amaury Hayat, Gabriel Ebner, Aurélien Rodriguez, and Timothée Lacroix. HyperTree Proof Search for Neural Theorem Proving. arXiv:2205.11491, 2022.

[26] Nayoung Lee, Ziyang Cai, Avi Schwarzschild, Kangwook Lee, and Dimitris Papailiopoulos. Self-Improving Transformers Overcome Easy-to-Hard and Length Generalization Challenges. arXiv:2502.01612, 2025.

[27] Yuhao Li and Shengchao Liu. On Distinguishing Capability Elicitation from Capability Creation in Post-Training: A Free-Energy Perspective. arXiv:2605.08368, 2026.

[28] Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models. arXiv:2505.24864, 2025.

[29] Alireza Mousavi-Hosseini and Murat A. Erdogdu. Post-Training with Policy Gradients: Optimality and the Base Model Barrier. arXiv:2603.06957, 2026.

[30] Maya Okawa, Ekdeep Singh Lubana, Robert P. Dick, and Hidenori Tanaka. Compositional abilities emerge multiplicatively: Exploring diffusion models on a synthetic task. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2310.09336.

[31] Simon Park, Simran Kaur, and Sanjeev Arora. How Does RL Post-training Induce Skill Composition? A Case Study on Countdown. arXiv:2512.01775, 2025.

[32] Gabriel Poesia, David Broman, Nick Haber, and Noah D. Goodman. Learning Formal Mathematics From Intrinsic Motivation. arXiv:2407.00695, 2024.

[33] Stanislas Polu and Ilya Sutskever. Generative Language Modeling for Automated Theorem Proving. arXiv:2009.03393, 2020.

[34] Stanislas Polu, Jesse Michael Han, Kunhao Zheng, Mantas Baksys, Igor Babuschkin, and Ilya Sutskever. Formal Mathematics Statement Curriculum Learning. arXiv:2202.01344, 2022.

[35] Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking. arXiv:2402.14811, 2024.

[36] Yuxiao Qu, Amrith Setlur, Virginia Smith, Ruslan Salakhutdinov, and Aviral Kumar. POPE: Learning to Reason on Hard Problems via Privileged On-Policy Exploration. arXiv:2601.18779, 2026.

[37] Yuval Ran-Milo, Yotam Alexander, Shahar Mendel, and Nadav Cohen. Outcome-Based RL Provably Leads Transformers to Reason, but Only With the Right Data. arXiv:2601.15158, 2026.

[38] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are Emergent Abilities of Large Language Models a Mirage? arXiv:2304.15004, 2023.

[39] Rylan Schaeffer, Noam Levi, Brando Miranda, and Sanmi Koyejo. Pretraining Scaling Laws for Generative Evaluations of Language Models. arXiv:2509.24012, 2025.

[40] Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, Yulia Tsvetkov, Hannaneh Hajishirzi, Pang Wei Koh, and Luke Zettlemoyer. Spurious Rewards: Rethinking Training Signals in RLVR. arXiv:2506.10947, 2025.

[41] Jingyan Shen, Ang Li, Salman Rahman, Yifan Sun, Micah Goldblum, Matus Telgarsky, and Pavel Izmailov. Understanding Reasoning from Pretraining to Post-Training. arXiv:2607.16097, 2026.

[42] Idan Shenfeld, Jyothish Pari, and Pulkit Agrawal. RL's Razor: Why Online Reinforcement Learning Forgets Less. arXiv:2509.04259, 2025.

[43] Avi Singh, John D. Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, et al. Beyond human data: Scaling self-training for problem-solving with language models, 2023.

[44] Wenhe Sun, Cunxiang Wang, Zijun Yao, and Yixin Cao. From Base Rollouts to RL Reasoning: A Budgeted Search Perspective. arXiv:2609.01274, 2026.

[45] Abiel Malkani Talwar, Christopher Nemeth, and Henry Moss. A Critical Look at Power Sampling when Test Time Sharpening LLMs. arXiv:2610.13653, 2026.

[46] Nikolaos Tsilivis, Eran Malach, Karen Ullrich, and Julia Kempe. How Reinforcement Learning After Next-Token Prediction Facilitates Learning. arXiv:2510.11495, 2025.

[47] Teun van der Weij, Felix Hofstätter, Oliver Jaffe, Samuel F. Brown, and Francis Rhys Ward. AI Sandbagging: Language Models can Strategically Underperform on Evaluations. arXiv:2406.07358, 2024.

[48] Siwei Wang, Yifei Shen, Haoran Sun, Shi Feng, Shang-Hua Teng, Li Dong, Yaru Hao, and Wei Chen. Benefits and Pitfalls of Reinforcement Learning for Language Model Planning: A Theoretical Perspective. arXiv:2509.22613, 2025.

[49] Gabriel Wu and Jacob Hilton. Estimating the Probabilities of Rare Outputs in Language Models. arXiv:2410.13211, 2024.

[50] Mingqi Wu, Zhihao Zhang, Qiaole Dong, Zhiheng Xi, Jun Zhao, Senjie Jin, Xiaoran Fan, Yuhao Zhou, Huijie Lv, Ming Zhang, Yanwei Fu, Qin Liu, Songyang Zhang, and Qi Zhang. Reasoning or Memorization? Unreliable Results of Reinforcement Learning Due to Data Contamination. arXiv:2507.10532, 2025.

[51] Zijian Wu, Suozhi Huang, Zhejian Zhou, Huaiyuan Ying, Zheng Yuan, Wenwei Zhang, Dahua Lin, and Kai Chen. InternLM2.5-StepProver: Advancing Automated Theorem Proving via Critic-Guided Search. arXiv:2410.15700, 2024.

[52] Tengyang Xie, Dylan J. Foster, Akshay Krishnamurthy, Corby Rosset, Ahmed Awadallah, and Alexander Rakhlin. Exploratory Preference Optimization: Harnessing Implicit Q*-Approximation for Sample-Efficient RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

[53] Huajian Xin, Z. Z. Ren, Junxiao Song, Zhihong Shao, Wanjia Zhao, Haocheng Wang, Bo Liu, Liyue Zhang, Xuan Lu, Qiushi Du, Wenjun Gao, Qihao Zhu, Dejian Yang, Zhibin Gou, Z. F. Wu, Fuli Luo, and Chong Ruan. DeepSeek-Prover-V1.5: Harnessing Proof Assistant Feedback for Reinforcement Learning and Monte-Carlo Tree Search. arXiv:2408.08152, 2024.

[54] Ran Xin, Chenguang Xi, Jie Yang, Feng Chen, Hang Wu, Xia Xiao, Yifan Sun, Shen Zheng, and Kai Shen. BFS-Prover: Scalable Best-First Tree Search for LLM-based Automatic Theorem Proving. arXiv:2502.03438, 2025.

[55] Chen Yang and Jun Chen. RLVR is a Kernel, Not a Function: Statistical Inference for pass@k Crossovers. arXiv:2609.22547, 2026.

[56] Haohui Yang, Jiaxing Sun, and Xiujun Ma. More Correct Mass, Worse Answers: Why Power Sampling Can Fail and How to Fix It. arXiv:2608.14420, 2026.

[57] Xinhao Yao, Lu Yu, Xiaolin Hu, Fengwei Teng, Qing Cui, Jun Zhou, and Yong Liu. The Debate on RLVR Reasoning Capability Boundary: Shrinkage, Expansion, or Both? A Two-Stage Dynamic View. arXiv:2510.04028, 2025.

[58] Lifan Yuan, Weize Chen, Yuchen Zhang, Ganqu Cui, Hanbin Wang, Ziming You, Ning Ding, Zhiyuan Liu, Maosong Sun, and Hao Peng. From $f(x)$ and $g(x)$ to $f(g(x))$: LLMs Learn New Skills in RL by Composing Old Ones. arXiv:2509.25123, 2025.

[59] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? arXiv:2504.13837, 2025.

[60] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. STaR: Bootstrapping reasoning with reasoning, 2022.

[61] Charlie Zhang, Graham Neubig, and Xiang Yue. On the Interplay of Pre-Training, Mid-Training, and RL on Reasoning Language Models. arXiv:2512.07783, 2025.

[62] Rosie Zhao, Alexandru Meterez, Sham Kakade, Cengiz Pehlevan, Samy Jelassi, and Eran Malach. Echo Chamber: RL Post-training Amplifies Behaviors Learned in Pretraining. arXiv:2504.07912, 2025.

[63] Xinyu Zhu, Mengzhou Xia, Zhepei Wei, Wei-Lin Chen, Danqi Chen, and Yu Meng. The Surprising Effectiveness of Negative Reinforcement in LLM Reasoning. arXiv:2506.01347, 2025.

A Glossary

ladder; rN; r0
one 16-round expert-iteration run from one pretrained seed; the model after N rounds; the pretrained base.

evaluation; x0, x1; guided
$k = 256$ independent attempts per theorem at $T = 0.8$; two sample seeds; the same with logically rejected steps redrawn (at most 10 per attempt).

σ, σ^+, lemma route, free positive reductio, classical-only, strict
see Table 1.

acquired
at least 50% of a family's strict held-out members proved in one evaluation.

within reach / not reached / certified at K
$\hat{p} \geq 1/K$; 0 successes in $\geq K$ attempts; one-sided 95% Clopper–Pearson bound $< 0.05/K$ (§2).

Δ_{cov}
theorems RL proves at $k = 256$ minus theorems the base reaches within K.

$\mathcal{D}$, $\mathcal{D}^-$, $\mathcal{D}^{1\%}$
the full, knockout and 1% dose pretraining sets (155,000 records each).

B Construction of the knockout set

$\mathcal{D}^-$ removes every $\mathcal{D}$ record whose written or pruned proof applies `DN` to a derived double negation: 16,961 records, of which 16,703 are σ and 258 cite an $\to$-elimination, $\vee$-elimination, $\bot$-elimination, `DN` or $\wedge$-elimination line. It refills 16,961 records drawn uniformly, length by length, from $\mathcal{D}$'s own unused source pools: lengths up to 6 from the depth-3 generator pool and 7–12 from the long-proof pool. Every refill record passes every $\mathcal{D}$ filter (no box depth ≥ 3; `DN` only on assumptions or premises) and contains no σ^+ and no lemma route. Under an order-sensitive and a premise-order-invariant renaming key, the refill shares no class with $\mathcal{D}$ or with any of the 100 data and target files of the repository. One of $\mathcal{D}$'s 11 exclusion pools was absent when the set was built and could not be checked. Every refill proof, translated to Lean, is accepted (16,961/16,961). Unchanged proofs

pass the same harness and goal-negated proofs fail it. Per-length counts are identical to $\mathcal{D}$'s. The dose set $\mathcal{D}^{1\%}$ swaps 1,550 refill records for removed σ records at $\mathcal{D}$'s σ length mix. A 0.1% set is built but unused.

C When each ladder first trained on each family

First round in which at least half of a family's strict training theorems had an accepted proof in the ladder (every training theorem receives 32 attempts per round; "–" = not within 16 rounds).

ladder	conv. contrap.	neg. cond.	Peirce	Peirce seq.	excl. middle	De Morgan
FULL s0/s1/s2	1/2/1	4/13/3	–/–/3	–/–/5	–/14/–	never
FULL s3/s4/s5	2/1/1	2/–/–	8/–/–	3/–/16	never	never
KNOCKOUT s0–s5	never	never	never	never	never	never
DOSE s0/s1/s2	2/1/3	3/7/7	12/–/–	7/–/–	never	never

D Budgets at r16, the number of evaluation theorems, and the estimator

At r16 the five allocations for full seeds s0/s1/s2 are $K_{\text{att}} = 7{,}147$, $K_{\text{train}} = 17{,}531/15{,}884/20{,}422$, $K_{\text{uni}} = 24{,}619/22{,}827/29{,}510$, $K_{\text{pub}} = 47{,}962/49{,}428/50{,}447$ and $K_{\text{adapt}} = 109{,}875/100{,}929/119{,}575$ base attempts per theorem. K_{adapt} exceeds the 66,304 attempts most hard theorems received, so the r16 sets at K_{adapt} are mostly undetermined.

Every K scales as $1/N_{\text{eval}}$. For the knockout certificates (Table 6), spreading each ladder's training-phase compute over the 106 strict theorems instead of the 322 evaluation theorems would triple K (seed 0: 70,846) and void seed 0's certificate. Spreading it over 1,054 theorems would cut K to 5,251–7,125; then every base but seed 1 certifies (5 of 6). The certified count is thus 0, 1 or 5 of 6 at $N_{\text{eval}} = 106$, 322 or 1,054.

Figure 9 shows the known-proof estimator's calibration. Because the ratios are binned by measured $\hat{p}$, they condition on at least one success, which inflates $\hat{p}$ in the lowest bins.

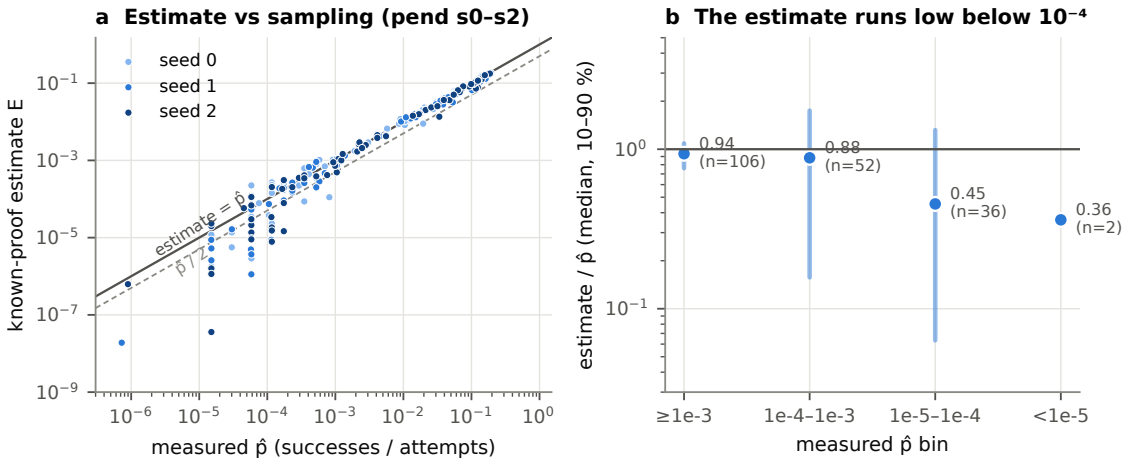


Figure 9: Known-proof estimator, full-exposure bases s0–s2. (a) Estimate against measured $\hat{p}$ for (seed, theorem) pairs with at least one success in large-k sampling. (b) Median and 10–90% range of $E/\hat{p}$ by $\hat{p}$ bin: 0.937 (106 pairs), 0.883 (52), 0.454 (36) and 0.36 (2).

Other measurements on the full seeds. At matched gains in item-response-theory ability, EI flags no more items as differentially improved than replay-only training. Eight RL rounds move the weights about as far as replay-only training (relative Frobenius norm 0.28 vs 0.30, one seed). Twelve operational definitions of "created" agree poorly: the mean pairwise Jaccard similarity of their sets is 0.29 at r8 and 0.34 at r16.

E Two more knockout-base proofs in Lean

Both are large-k successes of knockout bases before any RL. Lean 4.35.0 accepts each and rejects it with the goal negated (`paper/scripts/lean_examples.py`); the experiment's loop used Lean 4.34.1. (a) Knockout seed 1's base on `la_transfer_1246`, a held-out converse-contraposition member: σ, then `DN` in both branches of an $\vee$-elimination.

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬((S  R)  R)))) : (((S  R)  R) → S) := by
  have n1 : ((¬S) → (¬((S  R)  R))) := h1
  have n12 : (((S  R)  R) → S) := (fun (n2 : ((S  R)  R)) => by
    have n6 : (¬(¬S)) := (fun (n3 : (¬S)) => by
      have n4 : (¬((S  R)  R)) := n1 n3
      have n5 : False := n4 n2
      exact (n5 : False))
    have n11 : S := Or.elim n2 (fun (n7 : (S  R)) => by
      have n8 : S := Classical.byContradiction (fun hh => n6 hh)
      exact n8) (fun (n9 : R) => by
      have n10 : S := Classical.byContradiction (fun hh => n6 hh)
      exact n10)
    exact n11)
  exact n12
```

(b) Knockout seed 0's base on `la_transfer_916` at $T = 1$: the refutation is built by $\to$-introduction (`n6 : ¬S → False`), not $\neg$-introduction, which Lean accepts because $\neg\neg S$ unfolds to $\neg S \to$ False. No pretraining set contains `DN` on such a line.

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬P))) (h2 : P) : (R → S) := by
  have n1 : ((¬S) → (¬P)) := h1
  have n2 : P := h2
  have n6 : ((¬S) → False) := (fun (n3 : (¬S)) => by
    have n4 : (¬P) := n1 n3
    have n5 : False := n4 n2
    exact n5)
  have n9 : (R → S) := (fun (n7 : R) => by
    have n8 : S := Classical.byContradiction (fun hh => n6 hh)
    exact n8)
  exact n9
```

F Power sampling

The sampler targets p^α over whole proofs by Metropolis–Hastings with suffix resampling: 32 chains per theorem, blocks of 4 proof steps, 8 MH steps per block (32 in a convergence check), $\alpha \in \{4, 2\}$ (and 1.5 in the control). Acceptance rates were 0.55–0.74. On a development pilot the per-chain success was 0.752 at 8 MH steps per block and 0.760 at 32. The plain comparators received exactly the decoded tokens of the power runs and took 1.6–2.7 times less wall-clock. They kept no proof text, so their 1–2 successes could not be re-checked in Lean.

G Pre-registered expectations against outcomes

The experiment's pre-registration stated numeric expectations. Tallied independently, they *hit* on:

- the shape of $\mathcal{D}^-$;
- the general-ability gap on held-out greedy decoding and on non-σ theorems;
- 0/123 strict training theorems in round 1 on all knockout seeds, and 0/106 held-out strict theorems at r0 in the x0 evaluation (the x1 evaluation found 1 on seed 1, and the guided evaluation 1, 1, 0, 1, 3, 0 across seeds 0–5);
- knockout median log-probabilities at or below -15 nats;
- converse contraposition acquired on 3/3 new full seeds;
- at least one family on 6/6 full and 0/6 knockout seeds, the primary p and the arm medians;
- recovery of non-classical holdout250 by r16;
- $\Delta_{\text{cov}} > 0$ in both arms;
- the dose arm's round-1 σ proofs and its families;
- knockout seed 1's extension.

They *missed* on:

- the classical-only share falling by $\geq 70\%$ (observed 67.6%);
- textbook72 within ± 4 at r0 (observed -6.5 over all 72 theorems with six full seeds, two read at another batch; -4.0 on its 53 intuitionistic theorems with three);
- certificates on $\geq 5/6$ knockout bases (observed 2/6 on the 105 family members, 1/6 on the pre-registered 106);
- negated conditional on $\geq 2/3$ new full seeds (observed 1/3);
- the first σ^+ proof after round 4 (it came in round 1, on theorems supplying $\neg\neg G$);
- Δ_{cov} within $+8$ to $+35$ (two seeds at $+36$ and $+37$);
- 3–15 strict training theorems in the dose arm's round 1 (observed 2/17/2);
- a seeded-round gain on ≥ 3 seeds (observed 1/5).

The power-sampling expectations (0 solves) were met, but the sampler failed its positive control, so they are uninformative rather than hits; so is "0/106 for every sampler" (plain sampling at $T = 1$ solved one).

H Corrections applied

Three exploratory Boschloo p-values in the experiment's reports came from a transposed table. With the arms as columns, the RL-only test is $p = 0.0032$ (reported 0.0020), excluded middle 0.39 (0.30) and the three-versus-none family tests 0.034 (0.035). The primary $p = 0.00024$ and the dose test $p = 0.0033$ are unchanged. `paper/scripts/stats_extra.py` recomputes all of them. This paper also reports the knockout certificates on the pre-registered 106-theorem pool: the experiment's reports and their review used the 105 family members, which certifies seed 3 as well (Table 6). The review's ten rewordings are applied throughout. The most important replace "need" with "not acquired by this EI within 16 rounds", use "reach" only in its technical sense, and state the sharpening null for one sampler.

I This paper's checks and analyses

This writing run pre-registered its own checks (`preregistration/create-or-elicit-paper.md`, commit `c602e43c` and Addenda 1–4) before running them. The pod checks used two NVIDIA A40

instances. All evaluations are $k = 256$, $T = 0.8$, batch 512, on the 106 strict held-out theorems unless stated, with Lean in the loop.

C1: what the cut-offs hide. Paired evaluations with sample seed 0 on the same file, with standard caps (512 tokens per step, 96 steps) and doubled caps (1,024, 192). The draws differ only where a cap binds, apart from rare re-draws when the batch composition changes.

r16 checkpoint	proved, std → dbl	families acquired, std → dbl	cut-off share, std → dbl
knockout s0	0 → 0	none → none	0.04% → 0.00%
knockout s1	3 → 3	none → none	5.09% → 2.73%
knockout s2–s5	(pending)		
dose s2	28 → 29	cc, nc → cc, nc	9.30% → 8.58%
dose s0, s1; full s4	(pending)		

The pre-registered expectation that doubled caps would cut the cut-off share at least five-fold failed where cut-offs were large. Most cut attempts are non-terminating, and they stay cut at twice the budget. The counts barely move so far, so the cut-offs have not hidden successes that matter for any verdict.

C2: the dose arm on an equal footing. A second plain evaluation (sample seed 1) of dose r16 seeds 0, 1 and 2 proves 56, 39 and 30 strict theorems (first evaluation: 58, 39, 30) and acquires the same families on 3/3 seeds. The guided evaluation proves 58 and 42 on seeds 0 and 1 (seed 2 pending), again with the same families.

C3: textbook72 at the knockout's evaluation settings. The old full bases s0 and s1, evaluated at batch 512, prove 34 and 30 of textbook72's 72 problems, against 32 and 32 in their original batch-2,048 evaluation (seed 2 pending). Both are within the pre-registered ±4.

C4: knockout seed 1's members at $n \geq K$. (pending)

C5: the pre-filter, in full. (pending)

A2–A5: base prior, first-solve order and the waiting-time null. These are reported in §3.6 and §3.3, with their scripts in `paper/scripts/` (`figures.py prior`, `a3_member_prior.py`, `a4_first_solve.py`, `a5_waiting_time.py`). A2 was pre-registered after its inputs had been seen, and A5 was added after the first critique round; both are post hoc. A3 met its pre-registered expectation (AUC ≥ 0.7 on ≥ 3 of 4 full seeds: 4 of 4), as did A4 ($\rho \leq -0.5$ on 4 of 4).

J Provenance of earlier results

Run names refer to reviewed reports in the fork `dan-pandori/nd-takehome` (branch `dan_<run>`, file `review_<run>.md`) and to their summaries in the project's `experiment-summaries`.

- create-or-elicit: the knockout, dose and full s3–s5 ladders; the budgets and estimator calibration; power sampling. Review commits `dd8ee82e` (blind recount) and `8e195390` (verdict).
- capability-defs: the measurement case study on full seeds s0–s2 (budgets, certificates, controls, definitions, guided evaluations); the no-`DN` knockout taught by demonstrations (J6). Review `da11be78`.
- trajectory, rl-continue: full seeds s0–s2, rounds 1–8 and 9–16; likelihoods along RL.

- rl-from-ckpt: the replay-only control. guided-tts: guided redraws at matched tokens and the step checker. mcts-a: PUCT search with a learned value (review `5e70afbd`). grpo-best: GRPO variants from the same checkpoints.
- Earlier organisms (whole-proof token models of 3.2M–85M parameters, judged by the project's older checker with Lean confirmation where stated): the ignition study (review `review_ignition.md`), round 2 run 3 (minimum outside data), round 3 run 4a (reductio against model size) and the run4-grpo review.
- support-curves, support-state, claim-audit: the 3.2M interface result.

The claims ledger (`claims_ledger.md`, rows L1–L101) and `paper/LEDGER.md` map every inherited number to its reviewed source.